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Introduction

Overview

The purpose and scope of the analysis for this project is to successfully and effectively apply classification and clustering techniques to the dataset, extract insights, and recommend actionable strategies for the betterment of the business. Prior to that, data preparation, preprocessing, and exploratory data analysis will be done to build a solid foundation for advanced machine learning analysis. It involves verifying the data quality, understanding the dataset, and visualizing the data with various techniques such as bar charts and heatmaps. The project will also include training and evaluating the performance (through metrics) of different classification models, such as Random Forest, KNN, Decision Tree, and Logistic Regression for product categorization. On the other hand, clustering algorithms will be applied to group products and to identify the optimal number of clusters through silhouette and elbow methods. Overall, the key findings of these algorithms and models will be analyzed, compared, and interpreted in order to provide clear recommendations and conclusions for improving PriceRunner's product categorization, merchant performance, and sales strategies.

The PriceRunner dataset is highly relevant and ideal to NextPredict Solutions because of its vast amount of real-world product, category, and merchant data that can be used for classification, clustering, predictive analytics, and data-driven solutions. This will help NextPredict develop advanced machine-learning models that enhance merchant insights and optimize product classification, which helps businesses make data-driven decisions in a competitive market. For instance, this helps understand which categories and products are trending across the market or merchants through the use of classification and clustering techniques. It also enhances product searchability for greater customer satisfaction, while merchants can prevent misclassification errors to ensure that their products are categorized in the right group. The merchant data can be highly useful for identifying top-performing merchants across different categories and improving inventory management. Overall, the combination of all of these data analysis techniques will make PriceRunner a valuable resource for NextPredict's research and development efforts in improving product grouping algorithms, building a better recommendation system, and refining customer engagement strategies in a competitive market.

Data Analysis Language

The data analysis languages that are commonly used nowadays are R and Python, however, in this project, Python is primarily used due to its versatility, library ecosystem, and simplicity (Sidak, 2024). A data analysis language is designed to process, analyze, and visualize data and is equipped with certain functions, tools, and libraries. According to Eyre (2024), the following are Python's main capabilities and functions:

1. **Data Import and Export**

   Python allows importing and exporting datasets from several sources such as CSV files, Excel files, and SQL databases by installing and executing libraries like Pandas and Numpy. For example, importing a CSV file is written as follows:

```
import pandas as pd
data = pd.read_csv('data.csv')
```

   After analysis, Python will provide various formats for exporting the data, such as JSON, CSV, and Excel files.

```
data.to_csv('output.csv', index=False)
```

2. **Data Manipulation**

   Python allows for handling structured data in different ways, such as merging, grouping, filtering, and transforming large amounts of data efficiently through libraries like Pandas.

```
import pandas as pd

# Creating a sample DataFrame
data = {'Name': ['Alice', 'Bob', 'Charlie'],
        'Age': [25, 30, 35],
        'Salary': [50000, 60000, 70000]}

df = pd.DataFrame(data)

# Filtering data where Age is greater than 28
filtered_df = df[df['Age'] > 28]

print(filtered_df)
```

3. **Data Cleaning and Preprocessing**

   Python has the ability to handle and solve issues relating to missing values, inconsistencies, duplicates, and outliers in a

dataset with the assistance of NumPy and Pandas for cleaning and preparing data for analysis.

```
import pandas as pd
import numpy as np

# Creating a DataFrame with missing values
data = {'Name': ['Alice', 'Bob', 'Charlie', 'David'],
        'Age': [25, np.nan, 35, 40],
        'Salary': [50000, 60000, np.nan, 80000]}

df = pd.DataFrame(data)

# Filling missing values with mean
df.fillna(df.mean(numeric_only=True), inplace=True)

print(df)
```

4. Statistical Analysis

Python supports statistical analysis techniques like probability distributions, regression analysis, hypothesis testing, ANOVA, chi-square test, and more using SciPy and statsmodel.

```
import numpy as np
from scipy import stats

data = [10, 20, 30, 40, 50, 60]

# Mean and Median
mean_value = np.mean(data)
median_value = np.median(data)

# Linear Regression
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 6, 8, 10])
slope, intercept, _, _, _ = stats.linregress(x, y)

print(f"Mean: {mean_value}, Median: {median_value}")
print(f"Regression Line: y = {slope}x + {intercept}")
```

5. Data Visualization

Python enables users to develop visual representations of the data, such as bar graphs, scatterplots, box charts, histograms, and many more. This uses libraries like Seaborn, Plotly, Matplotlib,

```
import matplotlib.pyplot as plt
import seaborn as sns

# Sample data
data = {'Category': ['A', 'B', 'C', 'D'],
        'Values': [10, 25, 30, 15]}

df = pd.DataFrame(data)

# Creating a bar plot
sns.barplot(x='Category', y='Values', data=df)

plt.title("Category-wise Values")
plt.show()
```

6. Machine Learning & AI

Python allows predictive modeling and deep learning through machine learning and artificial intelligence libraries like PyTorch, scikit-learn, and TensorFlow.

```
from sklearn.linear_model import LinearRegression
import numpy as np

# Training data
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 6, 8, 10])

# Creating and training the model
model = LinearRegression()
model.fit(X, y)

# Predicting a new value
prediction = model.predict([[6]])

print(f"Predicted Value for X=6: {prediction[0]}")
```

7. Big Data Processing

Python interacts with large-scale data through big data frameworks like PySpark and ApacheSparke.

```
from pyspark.sql import SparkSession

# Initialize a Spark session
spark = SparkSession.builder.appName("BigDataExample").getOrCreate()

# Creating a DataFrame
data = [(1, 'Alice', 25), (2, 'Bob', 30), (3, 'Charlie', 35)]
columns = ['ID', 'Name', 'Age']
df = spark.createDataFrame(data, columns)

df.show()
```

8. Database Integration

Python has the ability to connect with several databases using SQLite and SQLAlchemy libraries in order to retrieve and fetch data.

```python
import sqlite3

# Connecting to database
conn = sqlite3.connect('example.db')
cursor = conn.cursor()

# Creating table
cursor.execute('''CREATE TABLE IF NOT EXISTS employees (id INTEGER, name TEXT, salary
REAL)''')

# Inserting data
cursor.execute("INSERT INTO employees VALUES (1, 'Alice', 50000)")
conn.commit()

# Fetching data
cursor.execute("SELECT * FROM employees")
print(cursor.fetchall())

conn.close()
```

General programming languages like C++, Java, and R can interact with Python as a data analysis language in many ways.

1. **Interaction with Java**

   Pyjnius is a Python library that enables the user to call Java classes and objects directly in the Python code. It can connect to an existing Python process (like Android) or begin a new Java Virtual Machine (JVM) (pyjnius 1.6.1 documentation, 2023). Similarly, **Jython** allows interaction with Python with Java, and it's particularly useful for experimenting with Java, embedding Python code in Java applications, and speeding up the development.

```python
from jnius import autoclass

Stack = autoclass('java.util.Stack')
stack = Stack()

stack.push('hello')
stack.push('world')
print(stack.pop())  # Outputs 'world'
print(stack.pop())  # Outputs 'hello'

$ python
```

   Similarly, **Jython** allows interaction with Python and Java, and it's particularly useful for the following tasks, available for non-commercial and commercial use, such as embedded scripting, interactive experimentation, and rapid application development (Jython, 2025).

```java
import org.python.util.PythonInterpreter;

public class JythonHelloWorld {
    public static void main(String[] args) {
        try(PythonInterpreter pyInterp = new PythonInterpreter()) {
            pyInterp.exec("print('Hello Python World!')");
        }
    }
}
```

2. **Interaction with R**

rpy2 is a Python package that enables users to run R from within Python. It creates an interface to begin an R session and allows interaction so that Python users can access the rich statistical and graphical capabilities of R while using Python programming language (Marsja, 2015).

```python
import rpy2.robjects as ro

# Assign a variable in R
ro.globalenv["x"] = 10

# Execute an R command
ro.r("y <- x + 5")

# Access the result in Python
y = ro.r("y")
print(y[0])  # Output: 15
```

3. **Interaction with C++**

Cython is an optimizing static compiler for Python, wherein it allows writing C extensions easily (Cython, 2015). In short, the user can utilize and interact with C or C++ while writing Python code. It also boosts performance by adding static type performance as if it's written in C. Cython provides a debugging approach for C and processes big complex data efficiently. Similarly, the **Python/C API** enables users to embed Python into C or C++ applications and write Python extension modules.

```
# example.pyx

# Import Python/C API headers
from cpython.pycapsule cimport *
from cpython.ref cimport Py_INCREF, Py_DECREF

# C function to be called from Cython
cdef extern from "example.c":
    PyObject* add_numbers(PyObject* self, PyObject* args)

# Cython wrapper function for Python
def cy_add_numbers(a, b):
    cdef PyObject* result
    result = add_numbers(None, (a, b))  # Call C function
    Py_INCREF(result)  # Increase reference count
    return result
```

Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that allows computers to learn from data, train on patterns, and make predictions without being explicitly programmed to do so (GeeksforGeeks, 2023). It involves creating models and algorithms that are capable of performing tasks that are similar to humans, such as analyzing data, categorizing images, and predicting price fluctuations. For example, in order for the computer to recognize images of dogs, the computer will be fed thousands of images of dogs and let the machine learning algorithm identify the common features and patterns that define a dog. Gradually, as the algorithm processes more images, it gets better or trained at recognizing dogs, even when presented with new data or images. Machine learning has played a significant role in different fields and industries by solving or predicting complex problems that are far off the scope of traditional programming (Crabtree, 2023). With that, it improves decision-making and handles intricate issues efficiently. 4 different types of machine learning that are commonly used today: Supervised, Unsupervised, Semi-supervised, and Reinforcement learning. The following are the uses of Machine Learning across industries (GeeksforGeeks, 2016):

1. It automates repetitive tasks with precision, such as email filtering (spam-free inbox), ML-powered chatbots (tracking and password resets), and data processing, wherein it automates large-scale invoice analysis for business insights. This innovative automation reduces the manual effort, time, and error to complete daily tasks.
2. It is capable of processing and analyzing large amounts of data, allowing valuable insights to be extracted and real-time predictions to be made. Some of its use cases include fraud detection in financial transactions and the prediction of personalized feed recommendations on social media platforms from millions of interactions.
3. It also helps in creating the application for voice or speech recognition or virtual personal assistants (VPA). It will help people find and collect the information or data that has been asked over the voice and provide them the best answers. Some of these applications are Siri, Alexa, and Samsung Bixby
4. It assists in detecting crime or tracks unusual behavior of people like stumbling, napping on benches, and standing still for a long period. Then, it creates an automatic alert notification for the guards, who can help to avoid any problems.

5. There are medical projects that involve machine learning, such as breast cancer classification, Parkinson's Disease Classification, and Pneumonia detection, wherein it is performed by machine learning models that have more than 90% accuracy. Machine learning is used in predicting the type of disease or whether some disease is going to happen in the future to a person.

There are several types of machine learning problems, each has different characteristics and applications (Kumar, 2021; GeeksforGeeks, 2023):

Supervised Learning

1. **Classification** algorithms involve predicting categorical target variables wherein they represent discrete classes. Its goal is to accurately predict the class labels of new instances based on their features. It classifies spam and not spam emails, fruits into categories, genres of music or film, and web text classification. A real-life example is weather forecasting, wherein this observation requires the use of previous records (temperature, humidity, wind direction) in order to predict patterns of weather conditions. 2 types include Binary Classification and Multi-class Classification. Some of the algorithms are Logistic Regression, K-Nearest Neighbors, Gradient Boosting Classifiers, Random Forest, and Neural Networks.
2. **Regression** algorithms involve predicting continuous target variables or numerical values. Regression finds relationships between two variables or more to predict. It predicts the prices of houses (based on location, size, number of rooms), sales of a product, and population growth over time. This technique is known to be easily understandable and interpreted, robust to outliers, and able to handle linear relationships easily. However, it can be sensitive to multicollinearity and may not be suitable for highly complex relationships. Types of regression include Simple Linear, Multiple Linear, Polynomial, Ridge & Lasso, and Support Vector Regression.
3. **Ranking** algorithms involve sorting data based on their relevance to a query by using labeled datasets to train data and classify new data to predict outcomes. It is made up of 2 main factors, which are queries (input value) and documents (output value), wherein a function assigns a relevance score to each query-document pair. This is commonly used in search engines (Google web searches), target advertisements, and recommendation systems wherein large

companies apply these to maximize user engagement and experience. An algorithm includes Bipartite ranking (Bipartite Rankboost, Bipartite RankSVM).

Unsupervised Learning

1. **Clustering** algorithms involve categorizing the data points into similar clusters or groupings based on similar characteristics and then applying cluster analysis. Some of the common elements that influence cluster formation are the shortest distance and the density of data points. This technique is greatly useful in finding hidden patterns and pinpointing segments. However, it may be dependent on the choice of parameters, sensitivity to initial conditions, and difficulty in handling noisy data. Algorithms include K-Means Clustering, DBSCAN, Hierarchical Clustering, BIRCH, and Gaussian mixture models.
2. **Association rule** finds dependencies or correlations between one data item and another data item or different features in a dataset. In short, it is a phenomenon where particular variables co-occur together, however, it does not imply causality. Some applications of the association include market basket analysis, web usage mining, anomaly detection, healthcare, and social sciences. Market basket analysis involves identifying items that are frequently purchased together by customers and are commonly used in retail. For example, if a customer buys butter (antecedent or “if”), he most likely will also buy bread (consequent or “then”), which is why these products are stored on a shelf or near proximity. The metrics include support, confidence, and lift to help assess the strength and reliability of the association rules. Algorithms related to this rule are the Apriori Algorithm, Eclat Algorithm, and F-P Growth Algorithm.

Data Understanding and Proposal

Dataset Overview

The company dataset that will be explored in this project is from PriceRunner. PriceRunner is an independent price comparison service that helps shoppers find better products and cheaper prices in order to make smart decisions and save money (About PriceRunner, 2022). It was founded in 1999 and is active in Sweden, the UK, Denmark, and Norway, however, it also includes shipping from foreign countries. They have over 5.8 million products and 15.4 million visits every month in all countries. The dataset was collected from the UC Irvine Machine Learning. The UCI Machine Learning Repository is a collection of databases, data generators, and domain theories. It is widely used by the machine learning community, students, researchers, and educators to analyze ML algorithms and gather machine learning datasets (UCI Machine Learning Repository, 2019).

The dataset that will be used in the project is called *Product Classification and Clustering*, which contains 35,311 product offers or instances and consists of 10 categories provided by 306 different merchants. According to Akritidis (2020), this dataset is suitable for clustering, classification, entity-matching algorithms, and problems involving text mining in the context of product data. The dataset is in a tabular and text format, and it shows that there was previous data preprocessing performed, such as case folding and punctuation removal on the titles of column 2. It is stated under the variables table that there are no missing values on all 7 features in the dataset to ensure data integrity for analysis. The variables include Product ID, Product Title, Merchant ID, Cluster ID, Cluster Label, Category ID, and Category Label.

Data Dictionary

Variable Name	Role	Type	Description	Missing Values
Product ID	Feature	Integer	A unique numeric identifier is assigned to each product offer recorded by a merchant.	No
Product	Feature	Categorical	The product name or	No

Title			description, recorded by the merchant, which includes its brand name and specifications	
Merchant ID	Feature	Integer	A unique numeric identifier is assigned to each merchant selling the product.	No
Cluster ID	Feature	Integer	A unique numeric identifier that groups similar product listings.	No
Cluster Label	Feature	Categorical	A standardized product name is assigned to a group of similar product listings.	No
Category ID	Feature	Integer	A unique 4-digit numeric code represents the product category to which the item belongs.	No
Category Label	Feature	Categorical	The descriptive name of the category that the product is classified under.	No

[8]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn import datasets, metrics
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report, accuracy_score
from sklearn.model_selection import cross_val_score
from sklearn.metrics import confusion_matrix
from sklearn.cluster import KMeans
from collections import Counter
import re
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import cross_val_score, StratifiedKFold
from imblearn.over_sampling import SMOTE
from sklearn.decomposition import PCA
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.metrics import confusion_matrix, roc_auc_score, roc_curve, classification_report, accuracy_score, ConfusionMatrixDisplay
from scipy.stats import ks_2samp
```

➢ The first imperative step in every data analysis process is to import the essential libraries (Seaborn, Pandas, Numpy, and Matplotlib) and modules that are generally used for data manipulation, visualization, general plotting, and

numerical calculations. Additionally, a library like ‘sklearn’ contains machine learning algorithms and metrics, including Logistic Regression, Random Forest, K-Neighbors, and more.

```
[11]: df = pd.read_csv("C:\\Users\\amand\\Downloads\\YEAR 2\\Term 2\\pricerunner_aggregate.csv")
```

➢ The ‘pd.read_csv’ function of Pandas will allow the program to read the PriceRunner dataset or the CSV file.

```
In [15]:
df.head()
```

Out[15]:

	Product ID	Product Title	Merchant ID	Cluster ID	Cluster Label	Category ID	Category Label
0	1	apple iphone 8 plus 64gb silver	1	1	Apple iPhone 8 Plus 64GB	2612	Mobile Phones
1	2	apple iphone 8 plus 64 gb spacegrau	2	1	Apple iPhone 8 Plus 64GB	2612	Mobile Phones
2	3	apple mq8n2b/a iphone 8 plus 64gb 5.5 12mp sim...	3	1	Apple iPhone 8 Plus 64GB	2612	Mobile Phones
3	4	apple iphone 8 plus 64gb space grey	4	1	Apple iPhone 8 Plus 64GB	2612	Mobile Phones
4	5	apple iphone 8 plus gold 5.5 64gb 4g unlocked ...	5	1	Apple iPhone 8 Plus 64GB	2612	Mobile Phones

➢ To get a quick glance at the dataset, the ‘.head()’ method of Pandas is used. It is observed that each row in the data contains the product offering, along with its merchant provider and the category it belongs to. Initially, it is seen that the dataset includes mobile phones and other types of consumer electronics. Each product offering is assigned a unique ‘Product ID’, while ‘Merchant ID’, ‘Cluster ID’, and ‘Category ID’ depend on the product grouping and which merchant is associated with the product. This dataset is considered to be product-related data that can be used for further data analysis.

```
In [23]: df.isnull().sum()

Out[23]: Product ID        0
         Product Title     0
         Merchant ID       0
         Cluster ID        0
         Cluster Label     0
         Category ID       0
         Category Label    0
         dtype: int64
```

➢ The ‘.isnull()’ with the ‘.sum()’ method confirms that none of the columns have missing values, as also previously mentioned in the data dictionary.

```
In [27]: duplicates = df.duplicated().sum()
         print(f"\nNumber of duplicate rows: {duplicates}")

Number of duplicate rows: 0
```

➢ The '.duplicated()' with the 'sum()' method checks the total number of duplicated rows. In this dataset, there are no duplicated rows. Therefore, the PriceRunner dataset is now well-prepared for data analysis, given there are no missing values or duplicated rows.

```
In [25]: df.columns = df.columns.str.strip()
         print(df.columns)

Index(['Product ID', 'Product Title', 'Merchant ID', 'Cluster ID',
       'Cluster Label', 'Category ID', 'Category Label'],
      dtype='object')
```

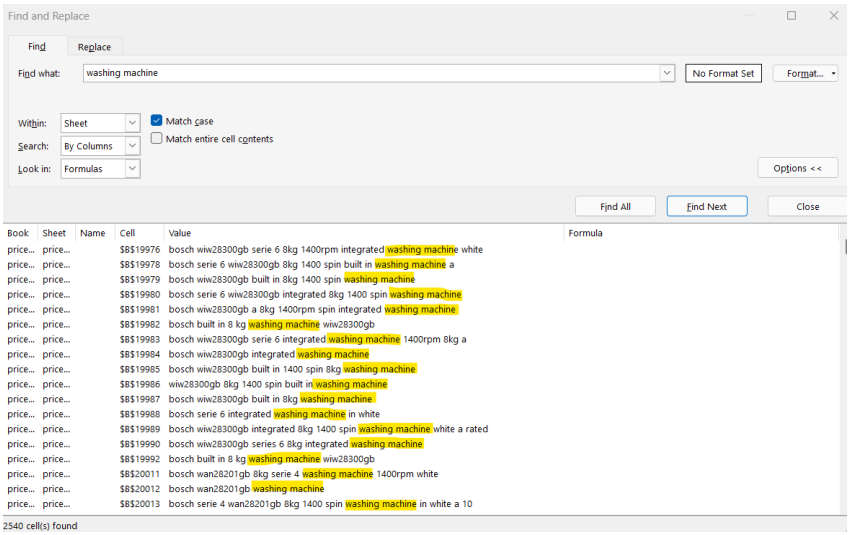
➢ Using the '.strip()' method removes the unnecessary space at the beginning of the columns to avoid random errors from white spaces in the later code. This standardizes the column names.

```
Top 5 most frequent values:
Product Title
washing machine                          90
built in fully integrated dishwasher     35
frost free fridge freezer                34
washer dryer                             24
american fridge freezer                  22
Name: count, dtype: int64

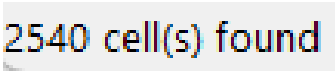
Top 5 most frequent values:
Cluster Label
Canon IXUS 185              27
Canon PowerShot SX730 HS    24
Samsung UE49NU7100          24
Apple iPhone 8 Plus 64GB    23
Samsung UE65NU7100          23
Name: count, dtype: int64

Top 5 most frequent values:
Category Label
Fridge Freezers     5501
Mobile Phones       4081
Washing Machines    4044
CPUs                3862
Fridges             3584
Name: count, dtype: int64
```

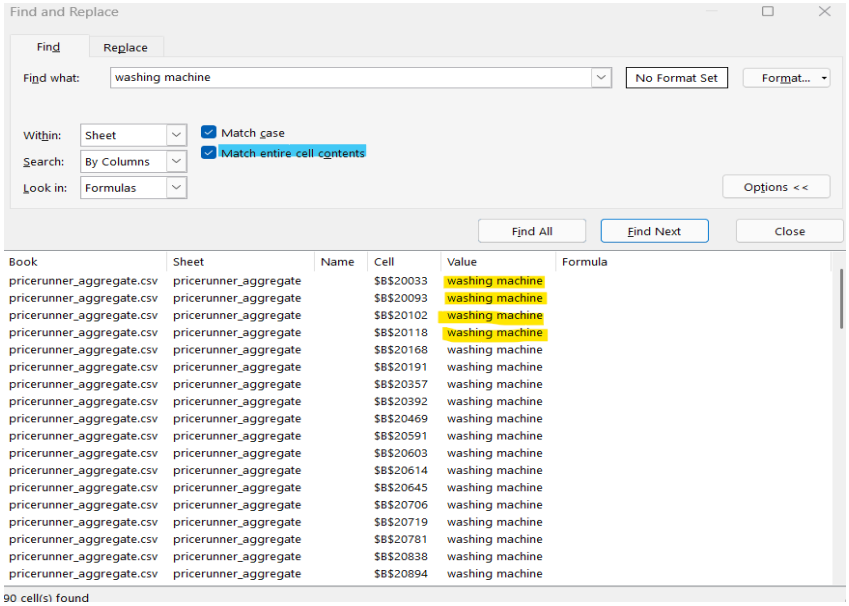
➢ Before proceeding to the data visualization, thorough cross-checking and observation were done upon acquiring these results from the code. While analyzing the most frequent values in the 'ProductTitle' column, it is observed that the highest frequency counts were very low (e.g. 95,35,34,24,22…) despite the dataset consisting of over 35,000 entries. This concerns the validity of these findings.

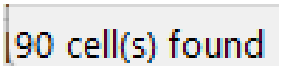


➢ To examine further, the dataset was manually reviewed in Microsoft Excel to confirm whether “washing machine” and other frequent values of product titles truly appeared as often as stated by the code output.



➢ Surprisingly, in Excel, it was found that 2,540 entries contain the phrase “washing machine” within the ‘ProductTitle’ column—far exceeding the 90 occurrences stated in the Jupyter Notebook. This discrepancy was due to how the program tries to match the entire cell contents rather than identifying occurrences within the product descriptions. For example, the entry “washing machine” is treated differently from a longer description entry like “bosch wan28201gb washing machine” or “Samsung washing machine 7kg”.





- This cross-checking examination revealed inconsistencies in the Product Title values, such as extra details, misspelled words, and variations in the order of words. The absence of standardized formatting made frequency analysis unreliable. Instead, using a Category Label or Cluster Label for analysis would yield more accurate insights. To address this issue, a 4-gram analysis and visualization were performed to capture multi-word patterns, which provide a more accurate representation of product descriptions. Unlike frequency analysis, which overemphasized generic terms like “washing machine”, the 4-gram approach aggregated important keyword clusters and contextualized product titles more effectively and validly. This adjustment led to a more improved and reliable visualization and analysis.

Proposal for Analysis

Overview	
Title of Analysis	It is important to create a title for the big data analysis in order to provide a quick context and clarify the focus for the audience or user, whether it be identifying trends or building predictive models.
Objectives	The objectives of the project must be defined in order to achieve the goal of the analysis. It should be aligned with the business problems or needs.
Data Overview	The analyst must lay out important details about the source of the dataset, size, data type, variables, and time span of data in order to establish credibility and transparency on the data.
Method of Analysis	
Data Cleaning and Preprocessing	One of the most crucial steps in data analysis is to remove missing, irrelevant, and duplicated data (GeeksforGeeks, 2024). It is also important to standardize formats like dates, addresses, and phone numbers in order to make the data easier to analyze and ensure consistency. Finding and addressing outliers must be done since it can distort analyses and lead to inaccurate results. Other data cleaning and preprocessing steps include fixing structural errors, converting data types, and language translation. Overall, data cleaning and preprocessing is an important step in order to ensure that the analysis is reliable, organized, accurate, and consistent.
Exploratory Data Analysis (EDA)	EDA helps understand the dataset by showing the summary statistics (e.g., mean, median, mode), shape, number of features, data type, and so on (GeeksforGeeks, 2021). Data visualization is included in this section, wherein the data points are illustrated in frequency tables, bar plots,

	histograms, scatter plots, and more. Prior to that, data transformation must be done, such as scaling numerical variables, encoding categorical variables, and aggregating data.
Data Segmentation	Data segmentation is the process of splitting the dataset into discreet groups based on attributes for comparison purposes (e.g., geographic location and demographic groups) (GeeksforGeeks, 2021). This step is important in improving model accuracy, increasing efficiency, and reducing model bias. Clustering techniques can be applied such as DBSCAN or k-means to identify groupings or clusters in the data.
Statistical Analysis	Statistical analysis is the systematic process of gathering, analyzing, and interpreting data using statistical methods such as hypothesis testing, correlation coefficients, regression models, t-tests, ANOVA, and many more (GeeksforGeeks, 2024). Statistical analysis can be descriptive, inferential, predictive, causal, and prescriptive. One can perform Natural Language Processing (NLP) or time series forecasting for advanced analysis, especially if there are large volumes of data (Staff, 2024).
Anticipated Outcomes	
The anticipated outcome of this analysis is to identify and uncover meaningful correlations and patterns through exploratory data analysis and data segmentation. It is also anticipated to have predictive models to forecast future events or trends with the use of statistical algorithms and machine learning techniques. This analysis will also provide recommendations and insights that can inform data-driven and strategic decisions such as customer segmentation, market opportunities, risk management, and other outcomes relating to the goal of the project. The machine learning models' performance must be evaluated using metrics like precision, recall, F1-score, and accuracy for classification tasks. Lastly, valuable insights will be integrated and presented through visualizations like dashboards and graphs for easier decision-making and interpretations.	

The Interaction between General Programming Languages and Data Analysis Languages

In this era where there are continuous technological advancements, efficiency and performance in tasks are important, especially for developers. A powerful technique is by linking programming languages together. There are several advantages and disadvantages of using multiple languages in developers' tasks:

Advantages	Disadvantages
It optimizes performance since a developer must choose the right and suitable programming language for the right task. For example, Python excels in rapid prototyping, data manipulation, and machine learning, while C++ is useful for tasks that require efficiency and speed.	It increases the complexity of the development process since each language has its own structure, paradigms, and syntax, making it difficult for the developers to maintain consistency in the codebase.
Each programming language has its own distinct features, making it a powerful tool for specific tasks. One must leverage its strengths to create a more dynamic project. For example, SQL is ideal for database management, R is powerful for statistical analysis, and JavaScript is great for developing interactive user interfaces.	It needs additional training, effort, and time investment to build knowledge in all the programming languages being used in the project to avoid errors or inefficiencies (learning curve).
It enables cleaner, manageable, and more organized code, especially for large projects. For example, it is recommended to use JavaScript for creating the user interface or dashboard and Python for processing backend data.	When the languages interact inefficiently, it may lead to performance issues or delays since data might require to be converted between systems in different languages.
It allows easier and more productive collaboration with specialists or developers since each one has its own expertise in multiple languages. Each developer will get to comfortably work on what language they know the best. For example, one may use HTML or CSS for front-end app development, while the back-end team works with Python or R.	As the project grows bigger, there may be a difficulty in maintenance wherein there's a requirement for understanding different debugging approaches, compatibility (e.g., frameworks or libraries), and paradigms involved in multiple languages.

To combine multiple languages, it is important to use Application Programming Interfaces (API). It enables different parts of the program to communicate with one another despite being written in several programming languages (Nigrange, 2024). Furthermore, it is common in a project to embed code from a different language directly inside them. For example, a Python script can contain SQL code to query a database, which combines SQL's querying power and Python's logic. Some tools to combine multiple programming languages are Docker, WebAssembly, and Jupyter Notebook. The following are the best languages to combine in a project (Nigrange, 2024; Miller, 2021):

1. **Python and R –** Python is mainly used for data collection and preprocessing. On the other hand, R is used for advanced statistical computing and data visualisation. The combination of these programming languages is suitable for data mining, bioinformatics, and data analysis.
2. **Javascript and Python** – Javascript is vastly used for front-web developments since it produces many dynamic and interactive user interfaces. On the other hand, Python is widely known for its ease of use and back-end services like data automation and processing. Therefore, JavaScript is recommended for building front-end, while Python is for managing back-end functions like database and data processing processes. This makes a great full-stack web application.
3. **C++ and Python** – C++ is known for its efficiency and speed, which is ideal for performance-critical applications such as simulations and game engines. Meanwhile, Python is much easier to write and is versatile, which is useful for scripting, logic, and rapid development. Therefore, C++ is handy for coding the core engine of a video game, while Python is much more suitable for less-performance tasks like game scripting and AI behavior.
4. **HTML/CSS and Javascript** – HTML/CSS provides the structure, layout, and design of the webpage, while Javascript is very valuable in interactive elements like animations, real-time content updates, and form submissions.

Modifying the Program with Both Data Analysis Language and General Programming Language

1. **SQL for Data Storage and Retrieval**

```
conn = sqlite3.connect("products.db")
cursor = conn.cursor()

# Create a table
cursor.execute('''
    CREATE TABLE IF NOT EXISTS products (
        product_id INTEGER PRIMARY KEY,
        product_title TEXT,
        merchant_id INTEGER,
        cluster_id INTEGER,
        cluster_label TEXT,
        category_id INTEGER,
        category_label TEXT
    )
''')

df.to_sql("products", conn, if_exists="replace", index=False)

print("Data stored in SQL successfully!")

df_sql = pd.read_sql_query("SELECT * FROM products", conn)
df_sql.head()
```

In the same environment, SQL was introduced to store and retrieve the dataset in an SQLite database to allow structured querying instead of reading a CSV file only. In short, Python retrieved data dynamically from SQL.

2. R for Statistical Analysis and Visualization (using rpy2 in Python)

```
pandas2ri.activate()

r_df = pandas2ri.py2rpy(df_sql)

robjects.r('''
    anova_test <- function(df) {
        result <- aov(category_id ~ cluster_id, data=df)
        return(summary(result))
    }
''')

anova_result = robjects.r['anova_test'](r_df)
print("ANOVA Test Results (from R):")
print(anova_result)
```

To analyze category trends, R is introduced to perform statistical tests like ANOVA in order to examine variance between clusters and identify statistical significance. In this case, Python calls R at runtime to execute ANOVA for statistical analysis.

```
ANOVA Test Results (from R):
               Df Sum Sq Mean Sq F value Pr(>F)
cluster_id      1 394523  394523  220154 <2e-16 ***
Residuals   35309  63275       2
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results show that Cluster ID is statistically significant since the p-value is <2e-16, which means there is a strong correlation between Cluster ID and Category ID. The F-value, 220154, is extremely high, which demonstrates an extreme effect of Cluster ID on Category ID.

```
pandas2ri.activate()

r_df = pandas2ri.py2rpy(df_sql)

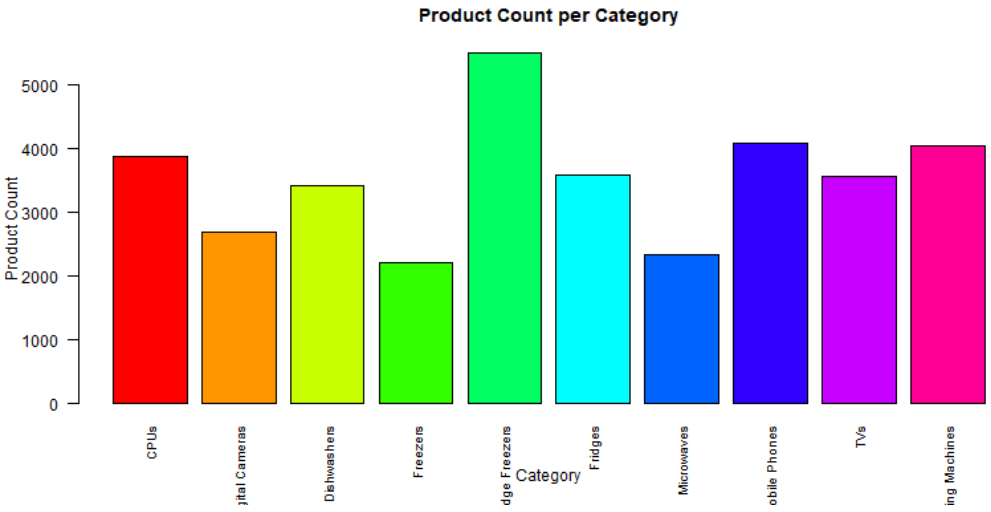
robjects.r('''
    plot_product_distribution <- function(df) {
        png("product_distribution.png", width=800, height=400)

        # Count occurrences of each category
        category_counts <- table(df$category_label)

        # Create bar plot
        barplot(category_counts, col=rainbow(length(category_counts)),
                xlab="Category", ylab="Product Count",
                main="Product Count per Category", las=2, cex.names=0.8)

        dev.off()
    }
''')

robjects.r['plot_product_distribution'](r_df)
print("✅ Bar chart saved as 'product_distribution.png'")
```



In base R, a bar chart has been created to understand the number of products belonging to each product category. This integrates R into Python using the rpy2 library, which enables Python to execute R functions, such as generating a bar chart from a Pandas DataFrame that is composed of categorical variables.

Conclusion

Combining general programming language with data analysis language is a strong approach for improving data management, performance optimization, and expanding the functionality of computationally intensive jobs. To create efficient

solutions, analysts can use Python for data retrieval and preprocessing, R for statistical analysis, and SQL for structured data storage and query. The project successfully executed how Python and R can be linked together at runtime with the use of the rpy2 library, which allows Python to call R for creating a bar chart and ANOVA statistical analysis. Moreover, the output of the ANOVA test resulted in a positive statistical significance between Category ID and Cluster ID, which proves the effectiveness of linking multiple languages together to analyze the data. Although it improves its overall functionality, it may present difficulties like learning curves, increased complexity, and performance overhead. However, with the use of APIs and integration tools (e.g., Docker, Jupyter Notebook, and WebAssembly), analysts can overcome these difficulties and create flexible and scalable machine learning models or applications.

Data Analysis and Machine Learning

Exploratory Data Analysis (EDA)

```
In [13]:
df.shape

Out[13]:
(35311, 7)
```

➢ The '.shape' attribute of Numpy shows the number of rows and columns, respectively. It displays that there are 35,311 rows and 7 columns in total.

```
In [17]:
df.dtypes

Out[17]:
Product ID          int64
Product Title      object
 Merchant ID        int64
 Cluster ID         int64
 Cluster Label     object
 Category ID        int64
 Category Label    object
dtype: object
```

➢ With the '.dtypes' object of Numpy, the data type of each column is shown. The columns 'Product ID', 'Merchant ID', 'Cluster ID', and 'Category ID' are 'int64' type, which means they store whole numbers. Meanwhile, 'Product Title', 'Cluster Label', and 'Category Label' are 'object' type that stores string or mixed data types.

```
In [19]:
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35311 entries, 0 to 35310
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Product ID      35311 non-null  int64
 1   Product Title   35311 non-null  object
 2    Merchant ID    35311 non-null  int64
 3    Cluster ID     35311 non-null  int64
 4    Cluster Label  35311 non-null  object
 5    Category ID    35311 non-null  int64
 6    Category Label 35311 non-null  object
dtypes: int64(4), object(3)
memory usage: 1.9+ MB
```

➢ The '.info' function of Pandas lays out a summary, including column name, non-null count, data type, and the total number of entries in the DataFrame. There are a total of 35,311 entries, with an index starting at 0. It is apparent that all of the columns do not have any missing values, which makes it advantageous for data analysis and eliminates the need for extensive data cleaning.

In [21]:
```
df.describe(include='all').T
```

Out[21]:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product ID	35311.0	NaN	NaN	NaN	26150.800176	13498.19122	1.0	14958.5	27614.0	37508.5	47358.0
Product Title	35311	30993	washing machine	90	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Merchant ID	35311.0	NaN	NaN	NaN	120.501883	117.045557	1.0	17.0	75.0	253.0	371.0
Cluster ID	35311.0	NaN	NaN	NaN	30110.687633	18410.265642	1.0	6123.0	40630.0	44059.0	47525.0
Cluster Label	35311	12849	Canon IXUS 185	27	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Category ID	35311.0	NaN	NaN	NaN	2618.14293	3.600708	2612.0	2615.0	2619.0	2622.0	2623.0
Category Label	35311	10	Fridge Freezers	5501	NaN	NaN	NaN	NaN	NaN	NaN	NaN

➢ The '.describe' method is utilized for providing summary statistics of the DataFrame. It reveals that there are 30,993 unique product titles, where 'washing machine' is the most frequent, appearing 90 times. The 'ProductID' would not be relevant to the analysis since it is a unique assignment for each product offering. The maximum value of 'MerchantID' is 371, which means the IDs range from 1 to 371, with each product associated with a merchant. The ClusterID ranges from 1 to 47,525. On the other hand, there are 12,849 unique 'ClusterLabel', with Canon IXUS 185 being the most frequent label, which appeared 27 times. The 'CategoryID' ranges closely from 2,612 to 2,623, which means that there are few categories, but they cover a large number of products. That said, the Fridge Freezers are the most frequent category, appearing 5,501 times, in the 'CategoryLabel'.

In [30]:
```
for i in df.columns:
    print(i,':',df[i].nunique())
```

```
Product ID : 35311
Product Title : 30993
Merchant ID : 306
Cluster ID : 13233
Cluster Label : 12849
Category ID : 10
Category Label : 10
```

➢ This code block displays the number of unique values in each column. Based on the findings, there are 306 unique merchants. There are also 10 unique 'CategoryID's, which matches the number of unique 'CategoryLabel'.

In [32]:
```
category_label_mapping = {label: idx for idx, label in enumerate(df['Category Label'].unique())}
print(category_label_mapping)
```

```
{'Mobile Phones': 0, 'TVs': 1, 'CPUs': 2, 'Digital Cameras': 3, 'Microwaves': 4, 'Dishwashers': 5, 'Washing M
achines': 6, 'Freezers': 7, 'Fridge Freezers': 8, 'Fridges': 9}
```

➢ To further expound on the 'Category Label' using mapping and a dictionary, it is observed that the categories contain values such as Mobile Phones, TVs, CPUs,

Digital Cameras, Microwaves, Dishwashers, Washing Machines, Freezers, Fridge Freezers, and Fridge, wherein the products are classified under one of these categories.

```
In [34]: categorical_cols = df.select_dtypes(include=['object']).columns
         print("\nCategorical Columns:")
         print(categorical_cols)

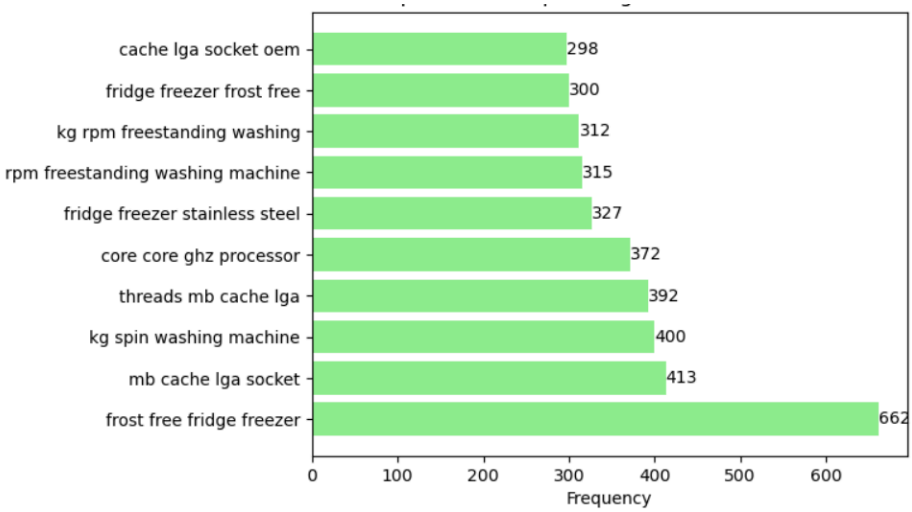
         numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
         print("\nNumerical Columns:")
         print(numerical_cols)

Categorical Columns:
Index(['Product Title', 'Cluster Label', 'Category Label'], dtype='object')

Numerical Columns:
Index(['Product ID', 'Merchant ID', 'Cluster ID', 'Category ID'], dtype='object')
```

- ➢ This code block identifies which features belong to categorical and numerical categories. This is critical for preprocessing data and feature selection when used for analyses and models later on.
- ➢ To improve the effectiveness of EDA, a key business question was formed: How can we improve product categorization, merchant performance, and overall sales strategy based on product titles, categories, clusters, and merchant behavior? To support this main question, various targeted sub-questions were formulated to guide the analysis and provide actionable insights.

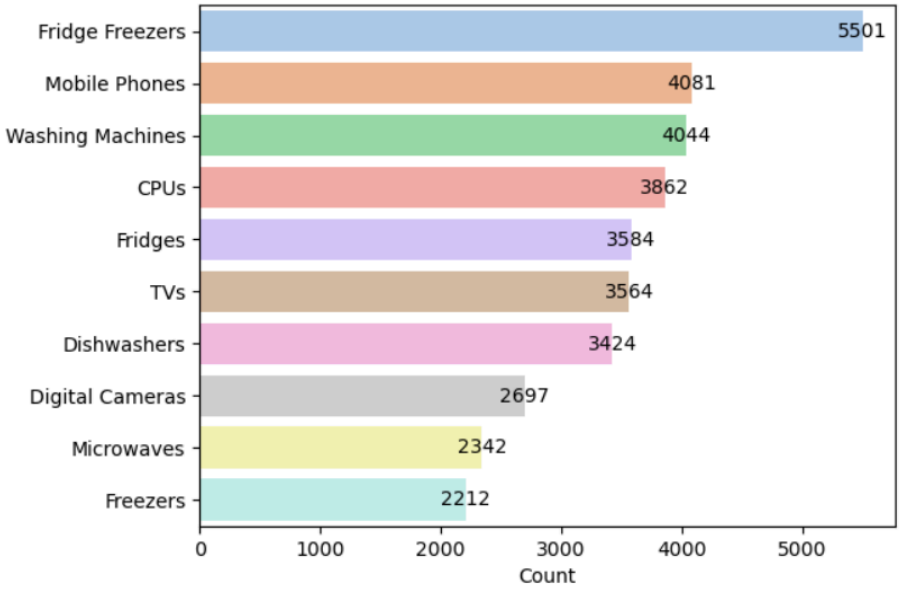
1. What are the most common 4-word combinations in product titles, and how can they help improve product categorization for the business?



It is evident that “frost-free fridge freezer” appeared most frequently in the dataset, with 662 occurrences. The following values are “mb cache lga socket”, “kg spin washing machine”, “threads mb cache lga”, and “core GHz processor”, wherein the counts are closer to each other. 4-grams can enhance product categorization, especially in identifying important product attributes. The 4-grams in Product Titles were used because they help identify the most frequent combinations of terms related to particular

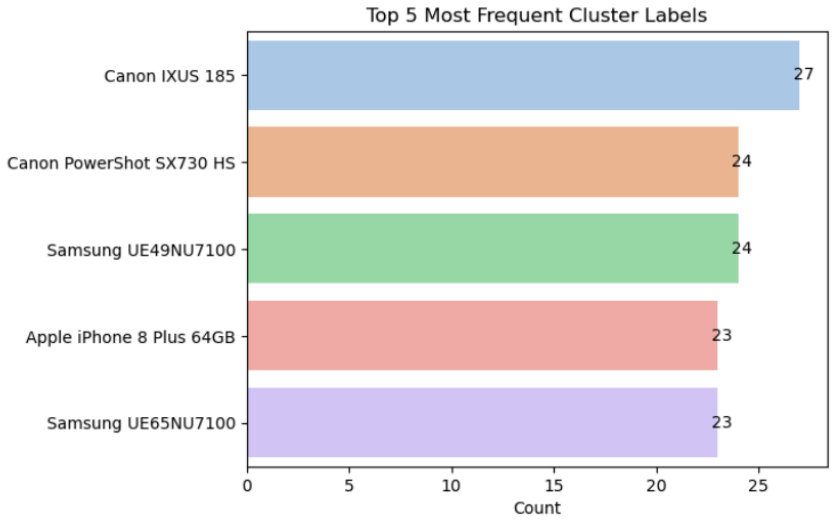
product features or types, which makes it easier to search. This is more effective than using the 'value.counts()' method since it provides inaccurate results. In this case, the most frequent product offerings were determined in the dataset with the use of keywords. Aside from categorization, 4 grams improve the searchability of the products. Overall, this method assists in identifying relevant products or facilitates product bundling to encourage upselling.

2. Which product categories have the most products, and how can we use this for optimization?



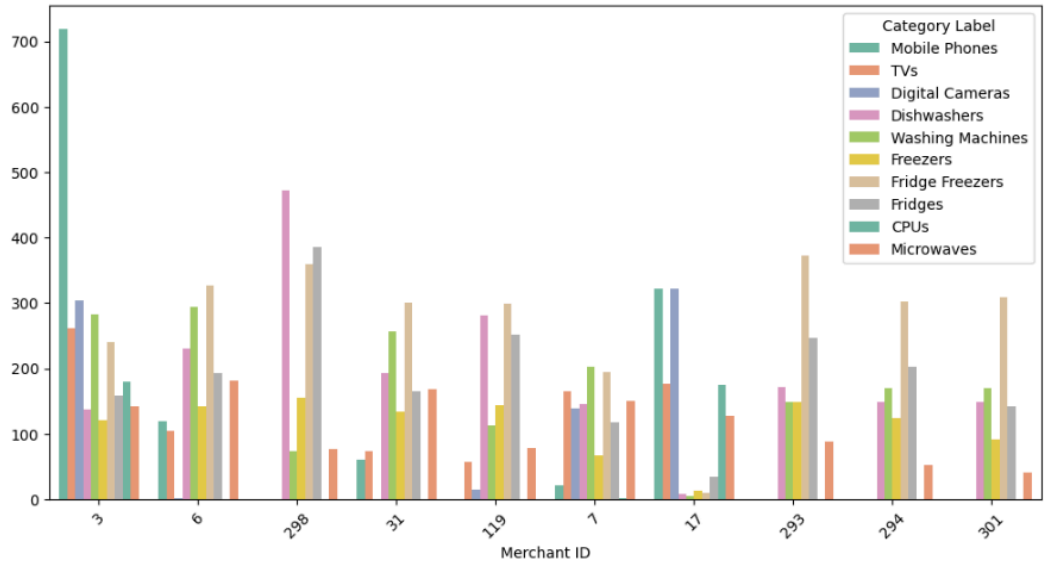
The product category with the highest number of products is Fridge Freezers, with a value of 5,501. It has a noticeable difference of more than 1,000 products compared to categories like Mobile Phones, Washing Machines, CPUs, Fridges, etc. The large number of fridge-freezer products available implies that the market may be oversaturated, which leads to intense competition with many options for customers to choose from. Therefore, the company can focus on advertising and promoting the top categories and highlighting these categories depending on the season or demand. The company can create great campaigns for this category, like "Top Fridge Freezers of 2025" or "Popular Mobile Phones under £1000". Since most of these are high-end products, it is best to improve the comparison features to a more detailed one wherein there are user reviews, ratings, and side-by-side comparisons of specifications. With that, customers can better navigate and understand the product offerings in the same category.

3. What are the most frequent product clusters, and how can clustering boost sales?



The most frequent product cluster is the Canon IXUS 185, with 27 occurrences, followed by the Canon PowerShot, Samsung UE49NU7100, Apple iPhone 8 Plus, and Samsung UE65NU7100, respectively. Canon models show high clustering frequencies, while Samsung TV models appear in two variants. Clustering is useful for product grouping because the company can put clustered products together in a bundle to create great deals for customers and increase sales. Clusters significantly help in grouping products by their brand, model, features, and price points. It can also improve the recommendation systems of the website by suggesting products within the same cluster to customers based on their searches, browsing, and purchasing history. Since the Canon IXUS 185, the company can design targeted marketing campaigns like advertising a cluster of Canon products with advertisements that showcase professional use features. Overall, this will enhance customers' shopping experience.

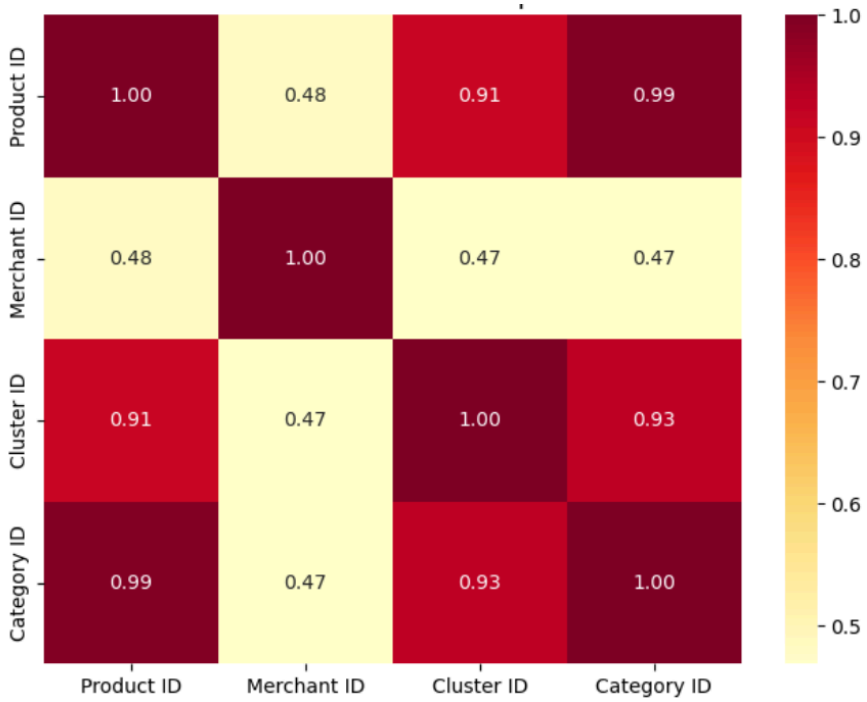
4. How do merchants perform across different categories, and how can this information optimize performance?




```
Merchant ID
3      2547
6      1591
298    1523
31     1350
119    1239
7      1204
17     1193
293    1177
294    1000
301     901
Name: count, dtype: int64
```

It is evident that the top-performing merchant is Merchant #3, with a total of 2,547 products. #3 stands out by concentrating on Mobile Phones, with over 700 products, while also offering products from different categories, which shows his/her diversity. Aside from #3, Merchant #298 is particularly strong in Dishwashers, with over 450 products. It is observed that Fridge Freezers, Freezers, and TVs have a consistent presence across several merchants. Additionally, #17 offers much more for Mobile Phones and Digital Cameras, which focuses on a tech-savvy customer base. The performance can be optimized when merchants focus on their strongest categories and continue to build inventory in these areas. The company can build strong relationships with top merchants and offer them better promotional support and marketing efforts. PriceRunner can also optimize the website interface to make it easy for the shopper to find and compare products of merchants like intuitive navigation and filter options.

5. Which variables show strong relationships with each other?



It is observed that Product ID and Category ID have a very strong correlation, nearly perfect at 0.99. This suggests that products are nearly aligned with their categories, which implies systematic product categorization. Cluster-ID and Category ID place the second-highest positive correlation (0.93), which indicates that clusters are greatly aligned with the product categories. This means that the clustering algorithms successfully grouped similar products. Lastly, the correlation of 0.91 between Cluster ID and Product ID shows that effective product clustering is based on similar characteristics.

Advanced Statistical Analysis

```
contingency_table = pd.crosstab(df['Category Label'], df['Cluster Label'])

chi2, p, dof, expected = chi2_contingency(contingency_table)

print(f"Chi-Square Statistic: {chi2}")
print(f"P-value: {p}")
```

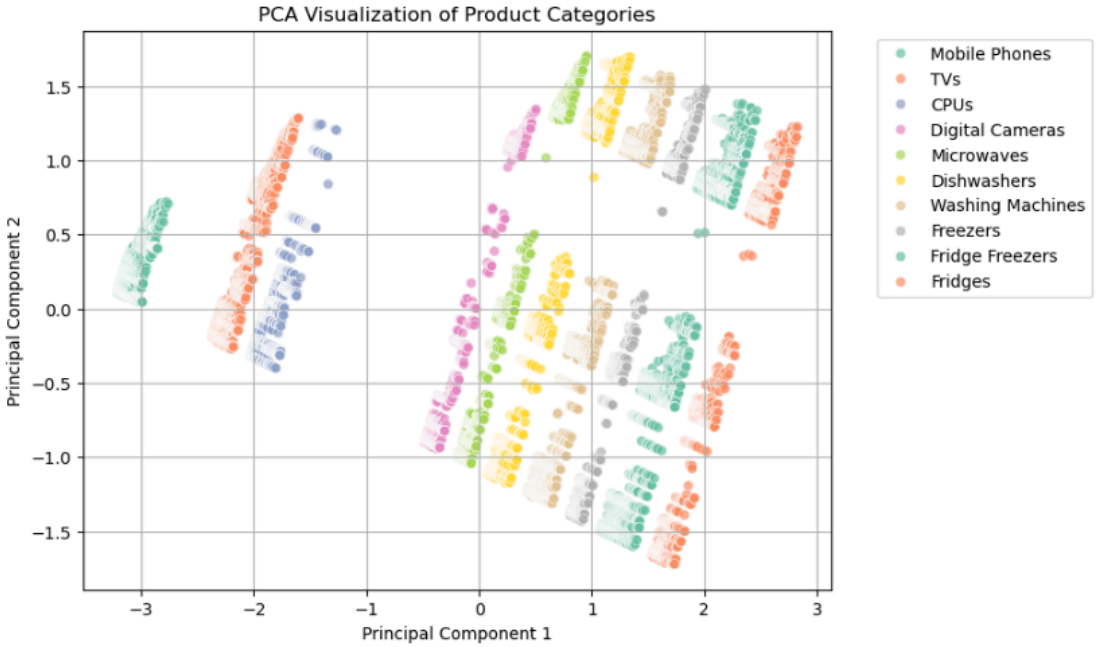
```
Chi-Square Statistic: 317714.5908326109
P-value: 0.0
```

The Chi-square test is a statistical method for categorical variables wherein it identifies the difference between observed and expected data (Biswal, 2021). A Chi-square test was performed to determine whether the Category Label is significantly correlated with the Cluster Label. The results show that there is a low p-value (<0.05), which means that these two categorical variables have a strong association with each other. A very large Chi-square value indicates that there is a high dependence between the variables, which may imply that the Category Label influences the distribution of Cluster Label.

```
categories = df['Category Label'].unique()
anova_results = f_oneway(*(df[df['Category Label'] == cat]['Product ID'] for cat in
categories))

print(f"F-statistic: {anova_results.statistic}, P-value: {anova_results.pvalue}")
```

One-way ANOVA is a statistical method to check if three or more independent groups are significantly different based on samples' means (Bobbitt, 2018). A one-way ANOVA analysis was done to check if there is a significant difference in the distribution of the Product IDs on different product categories. The low p-value (<0.05) shows that there is a highly significant difference, which means that the Product IDs are not evenly distributed among product categories, wherein some categories have more or fewer products. A very large F-statistic implies that there is a large variation between categories wherein particular categories dominated the dataset.



Principal Component Analysis (PCA) is a statistical method that is used for dimensionality reduction in large datasets and retains most of the original data (Jaadi, 2024). The PCA is tested on product categories, and it is evident that the points are well-separated and organized. This implies that the PC1 & PC2 effectively capture meaningful variances, which means that the dataset has a great relationship structure between product categories and numerical features. This suggests that the clustering techniques and classification models will perform well since there are already existing numerical features that divide the data successfully. Additionally, it most likely does not need extensive feature engineering like complex embeddings or polynomial features. The spaces in the PCA visualization suggest that the product categories are highly tied to numerical features like Cluster ID or Category ID. Overall, this demonstrates a good PCA.

Machine Learning Algorithms

1. **Supervised Learning –** It involves training algorithms on labeled datasets to predict outcomes or classify data accurately (GeeksforGeeks, 2017). This labeled data is already tagged with the correct answer. Then, the machine is provided with a new set of data so that the algorithm analyzes the training data and produces a correct classification or answer from the labeled data. In short, the trained machine can make predictions on new, unlabeled data. Most of the applications of supervised learning can solve a wide range of problems in real-world situations. This includes spam filtering, image classification, medical diagnosis, fraud detection, and natural language processing (NLP). Supervised learning consists of two categories of algorithms: regression and classification (GeeksforGeeks, 2017).
    - ➢ **Regression** – It is used to predict continuous values like stock prices, house prices, and customer churn. Some common regression algorithms are linear regression, polynomial regression, support vector machine regression, decision tree regression, and random forest regression.
    - ➢ **Classification** – It is utilized to predict categorical values like document classification, customer behavior prediction, and spam filtering. Some common classification algorithms include logistic regression, support vector machines, decision trees, random forests, and Naive Baye.

    It is important in supervised learning models to be evaluated in order to ensure that the model is generalizable and accurate (GeeksforGeeks, 2017). There are various different metrics that can be utilized for assessing the accuracy of these models. Starting off for regression models, metrics include Mean Square Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (Coefficient of Determination). On the other hand, classification includes metrics like accuracy, precision, recall, F1 score, and confusion matrix.

Advantages	Disadvantages
It allows data collection and produces output of data based on previous experiences.	Difficulty in classifying big data.
It optimizes performance criteria with the help of experience.	It requires a lot of computation time, therefore, more time and effort are allotted.

It is capable of performing classification and regression tasks	It is not capable of handling all complex tasks in machine learning.
An ability to take control over choosing the number of classes in the training data.	The training process is a requirement.
It allows the estimation of the result to a new sample.	A labeled data set is a requirement.

2. **Unsupervised Learning –** It involves a machine that learns from unlabeled data or does not have any pre-existing categories or labels (GeeksforGeeks, 2024). The aim of unsupervised learning is to uncover patterns and relationships in the data without any explicit guidance. Since there is no labeled data, the machine groups unsorted information to patterns, similarities, and differences without any prior training of data. Some real-world applications of unsupervised learning include anomaly detection, scientific discovery, recommendation systems, customer segmentation, and image analysis. Unsupervised learning consists of two categories of algorithms: clustering and association (GeeksforGeeks, 2024).
    - ➢ **Clustering** – It is used to group similar data points together by iteratively moving the points near to their cluster centers and far away from data points in other clusters. Examples of clustering include hierarchical clustering, k-means clustering, principal component analysis, singular value decomposition, and DBSCAN.
    - ➢ **Association Rule** – It is utilized to determine patterns in the data by finding relationships between different items in a dataset. Examples of association rule learning algorithms include the apriori algorithm, the Eclat algorithm, and the FP-Growth algorithm.

    It is important in unsupervised learning to evaluate the model in order to check if it is effective and useful (GeeksforGeeks, 2024). It can be more challenging since there is no actual, correct data to compare the model's predictions to. Some common metrics used include the silhouette score, Calinski-Harabsz score, Adjusted Rand index, Davies-Bouldin index, and F1 score.

Advantages	**Disadvantages**
Labeled training data is not a requirement.	Measuring effectiveness or accuracy can be challenging since it lacks predefined answers during the training phase.

It is capable of finding prior unknown patterns in the data.	The results frequently have lower accuracy.
Dimensionality reduction or PCA can be easily achieved.	It can be sensitive to data quality like outliers, missing values, and noisy data.
Great at finding patterns and relationships in data without being told what to look for.	When the model produces an output, the user frequently needs to spend time interpreting and labeling the classes that the model has identified, which can be extensive work.

3. **Reinforcement Learning –** It involves making decisions to maximize cumulative rewards in a given situation through experience. It tries to learn to achieve a goal in an uncertain environment by performing actions and gathering feedback through penalties or rewards. The learning process of this machine is characterized by different elements: policy, reward function, value function, and model of the environment. In reinforcement learning, the labels are given to sequences of dependent decisions. In other words, the output relies on the state of the current input, and the following input depends on the output of the previous output. Real-life applications of reinforcement learning include chess games, text summarization, robotics, and industrial control. The core concepts of reinforcement learning are the following:
    - Agent – The one who learns or performs decision-making.
    - Environment – The one the agent interacts with.
    - State – A particular situation in which the agent finds itself.
    - Action – All possible actions the agent can make.
    - Reward – Feedback in regard to the action taken from the environment.

    There are two types of reinforcement learning: positive and negative reinforcement.

    ➢ Positive – It is when an event occurs due to a certain behavior, which increases the frequency and strength of the behavior.
    ➢ Negative – It is when the behavior is strengthened due to a negative condition being stopped or strengthened.

Advantages	**Disadvantages**
It can be used to solve very complex problems that are unable	It is not ideal for solving simple problems.

to be solved with conventional techniques.	
During the training process, the errors can be corrected by the model.	It requires a lot of data and computation.
It has the ability to handle environments that are non-deterministic, which means that the outcomes of actions are not always predictable.	It highly relies on the quality of the reward function, wherein if it is designed poorly, the agent may not learn the desired behavior.
It can be combined with other machine learning techniques like deep learning in order to improve performance.	It is challenging to debug and interpret because the behavior of the agent is sometimes unclear.

Modelling

A training dataset teaches machine learning models to perform tasks or recognize patterns (GeeksforGeeks, 2023). This is used to identify the performance of the trained model, as well as to make decisions based on the previous cases and their results. The more training data that is fed to the model, the more knowledgeable and accurate predictions it can make over time. On the other hand, testing data is used to assess the model's performance, such as the progress and efficiency of algorithms. The testing data must be unseen or new and not part of the training data since it is "learned" by the model already. The purpose of separating training and testing data is to avoid overfitting and not simply memorizing the training data. This is to ensure that the model is making accurate predictions based on recent data. In short, training data is the learning phase, while testing data is the evaluation phase of the machine learning model. Splitting data for machine learning models is a critical step in developing one. The following are the common approaches for splitting the data into separate subsets: training, validation, and testing (GeeksforGeeks, 2020):

1. **Train-Test Split** – The dataset is separated into training and testing sets. The common data-splitting ratio is 80% training and 20% testing, whereas the training data must be larger.
2. **Train-Validation-Test Split** – The dataset is divided into three subsets: training, validation, and testing sets. The validation set is used to validate the model's performance during the training phase and to tune hyperparameters.
3. **K-fold Cross Validation** – The dataset is divided into k equally sized folds, and the model is trained and assessed k times.
4. **Stratified Sampling** – the distribution of crucial features or targets must be preserved in the training and testing sets, which is especially useful in dealing with imbalanced datasets.
5. **Time-based Split** – This is particularly used when dealing with time series data, where the training and testing sets are divided based on chronological order.

The target variable in the data analysis project is the Category Label, so it is best to use Label Encoder to transform categorical labels into numerical values to make it suitable for the machine learning model. The features were scaled using StandardScaler to standardize the data and improve the model performance.

```
le = LabelEncoder()
df['Category Label Encoded'] = le.fit_transform(df['Category Label'])

X = df.drop(columns=['Category Label', 'Category Label Encoded', 'Product ID', 'Product
Title'])
y = df['Category Label Encoded']

categorical_cols = X.select_dtypes(include=['object']).columns
for col in categorical_cols:
    X[col] = le.fit_transform(X[col])

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

This code block divided the dataset into a training set (X_train, y_train) and a testing set (X_test, y_test) using the 80:20 split ratio, which is the most common Train/Test split ratio. The Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training set in order to address the class imbalances in the dataset, which is common in developing machine learning models. This is to ensure that the model is prepared for evaluation.

```
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2,
random_state=20)

print(f"Training data shape: {X_train.shape}, Testing data shape: {X_test.shape}")

smote = SMOTE(random_state=20)
X_train_resampled, y_train_resampled = smote.fit_resample(X_train, y_train)
```

This will be tested into 4 different classification models, such as Random Forest, Decision Tree, KNN, and Logistic Regression, where each will produce an output of various classification metrics like accuracy, error rate, confusion matrix, Gini Coefficient, and K-S statistic to have a detailed report about the model's performances. However, in initial instances, the accuracy and classification report of the models were very high, and some had perfect scores (100% accuracy). A cross-validation was done to assess the model on different splits of data, but the models still performed exceptionally well. This means that the model predicted the categories of each data correctly all the time. It may not be bad in some cases since it may indicate that the data is well-prepared and there are few noise elements. However, it is possible that the model is overfitting, imbalanced, or easy-to-predict data. Therefore, it is highly important to verify the accuracy of the model.

```
Evaluating Random Forest Model
Accuracy of Random Forest: 100.00%
                  precision    recall  f1-score   support

            CPUs       1.00      1.00      1.00       778
 Digital Cameras       1.00      1.00      1.00       537
     Dishwashers       1.00      1.00      1.00       690
        Freezers       1.00      1.00      1.00       444
 Fridge Freezers       1.00      1.00      1.00      1081
         Fridges       1.00      1.00      1.00       693
      Microwaves       1.00      1.00      1.00       472
   Mobile Phones       1.00      1.00      1.00       859
             TVs       1.00      1.00      1.00       729
Washing Machines       1.00      1.00      1.00       780

        accuracy                           1.00      7063
       macro avg       1.00      1.00      1.00      7063
    weighted avg       1.00      1.00      1.00      7063
```

```
Evaluating K-Nearest Neighbors Model
Accuracy of K-Nearest Neighbors: 99.86%
                  precision    recall  f1-score   support

            CPUs       1.00      1.00      1.00       778
 Digital Cameras       1.00      1.00      1.00       537
     Dishwashers       1.00      1.00      1.00       690
        Freezers       1.00      0.99      0.99       444
 Fridge Freezers       1.00      1.00      1.00      1081
         Fridges       1.00      1.00      1.00       693
      Microwaves       1.00      1.00      1.00       472
   Mobile Phones       1.00      1.00      1.00       859
             TVs       1.00      1.00      1.00       729
Washing Machines       1.00      1.00      1.00       780

        accuracy                           1.00      7063
       macro avg       1.00      1.00      1.00      7063
    weighted avg       1.00      1.00      1.00      7063
```

```
Evaluating Decision Tree Model
Accuracy of Decision Tree: 100.00%
                  precision    recall  f1-score   support

            CPUs       1.00      1.00      1.00       778
 Digital Cameras       1.00      1.00      1.00       537
     Dishwashers       1.00      1.00      1.00       690
        Freezers       1.00      1.00      1.00       444
 Fridge Freezers       1.00      1.00      1.00      1081
         Fridges       1.00      1.00      1.00       693
      Microwaves       1.00      1.00      1.00       472
   Mobile Phones       1.00      1.00      1.00       859
             TVs       1.00      1.00      1.00       729
Washing Machines       1.00      1.00      1.00       780

        accuracy                           1.00      7063
       macro avg       1.00      1.00      1.00      7063
    weighted avg       1.00      1.00      1.00      7063
```

```
Evaluating Logistic Regression Model
Accuracy of Logistic Regression: 100.00%
                  precision    recall  f1-score   support

            CPUs       1.00      1.00      1.00       778
 Digital Cameras       1.00      1.00      1.00       537
     Dishwashers       1.00      1.00      1.00       690
        Freezers       1.00      1.00      1.00       444
 Fridge Freezers       1.00      1.00      1.00      1081
         Fridges       1.00      1.00      1.00       693
      Microwaves       1.00      1.00      1.00       472
   Mobile Phones       1.00      1.00      1.00       859
             TVs       1.00      1.00      1.00       729
Washing Machines       1.00      1.00      1.00       780

        accuracy                           1.00      7063
       macro avg       1.00      1.00      1.00      7063
    weighted avg       1.00      1.00      1.00      7063
```

```
Evaluating Random Forest Model with Cross-Validation
Cross-Validation Accuracy of Random Forest: 100.00%
Cross-Validation Scores: [100. 100. 100. 100. 100.]
Evaluating Decision Tree Model with Cross-Validation
Cross-Validation Accuracy of Decision Tree: 100.00%
Cross-Validation Scores: [100. 100. 100. 100. 100.]
Evaluating K-Nearest Neighbors Model with Cross-Validation
Cross-Validation Accuracy of K-Nearest Neighbors: 99.90%
Cross-Validation Scores: [99.86425339 99.90950226 99.88687783 99.89819005 99.95475113]
Evaluating Logistic Regression Model with Cross-Validation
Cross-Validation Accuracy of Logistic Regression: 100.00%
Cross-Validation Scores: [100. 100. 100. 100. 100.]
```

After extensive trial and error in testing the model, the problem has been identified and solved. To address the issue of very high accuracy and to improve the model's performance. The model's settings were configured, which made the results more realistic. It was found that the models' problems stemmed from class or data imbalance and mostly overfitting.

```
rf_model = RandomForestClassifier(n_estimators=5, max_depth=3, min_samples_split=10,
random_state=20, class_weight='balanced')
dt_model = DecisionTreeClassifier(max_depth=6, min_samples_split=10, random_state=20,
class_weight='balanced')
kn_model = KNeighborsClassifier(n_neighbors=20)
lr_model = LogisticRegression(class_weight='balanced', C=400, max_iter=500,
solver='liblinear')
```

Model Configurations:

1. **Random Forest and Decision Tree –** Reduced the n_estimators and max_depth to make the model more generalizable and prevent overfitting. Similarly, the min_samples_split was increased for the same reason.
2. **KNN** – Increased the default value (5) of n_neighbors to 20 because considering more neighbors generally smoothens the decision predictions of the model. Adding noisy data dropped the model accuracy to 54.37%. This is because KNN is a distance-based algorithm wherein if noisy data points are added, it significantly affects the predictions since the algorithm might incorrectly assign a noisy data point to the wrong class based on its proximity, which leads to lower classification accuracy. In short, the more noise is asses, the less the model generalizes the real patterns in the data since it relies upon incorrect data points. Therefore, it is normal for the KNN performance to decrease drastically when noisy data is introduced, mainly due to its sensitivity.
3. **Logistic Regression** - Increased the C parameter and reduced the max_iter parameter in order to prevent it from overfitting, making it a more generalized model.

```
def modeling(model_name, model, X_train, y_train, X_test, y_test):
    print(f"\nEvaluating {model_name} Model")
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    if hasattr(model, "predict_proba"):
        y_pred_proba = model.predict_proba(X_test)[:, 1]
    else:
        y_pred_proba = model.decision_function(X_test)

    accuracy = accuracy_score(y_test, y_pred) * 100
    error_rate = 100 - accuracy  # Calculate Error Rate

    print(f"Accuracy of {model_name}: {accuracy:.2f}%")
    print(f"Error Rate of {model_name}: {error_rate:.2f}%")

    print(f"Classification Report for {model_name}:\n{classification_report(y_test,
y_pred)}")

    cm = confusion_matrix(y_test, y_pred)
    print(f"Confusion Matrix for {model_name}:\n{cm}")

    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=model.classes_)
    disp.plot(cmap="viridis")
    plt.title(f"Confusion Matrix for {model_name}")
    plt.show()

    ks_statistic, ks_p_value = ks_2samp(y_pred_proba[y_test == 0], y_pred_proba[y_test ==
1])
    print(f"K-S Statistic for {model_name}: {ks_statistic:.4f}, p-value:
{ks_p_value:.4f}")

    gini_coefficient = 2 * ks_statistic - 1
    print(f"Gini Coefficient for {model_name}: {gini_coefficient:.4f}")

    return accuracy, error_rate, ks_statistic, gini_coefficient, y_pred
```

```
rf_accuracy, rf_error_rate, rf_ks_stat, rf_gini, rf_ypred = modeling("Random Forest",
rf_model, X_train_resampled, y_train_resampled, X_test, y_test)
dt_accuracy, dt_error_rate, dt_ks_stat, dt_gini, dt_ypred = modeling("Decision Tree",
dt_model, X_train_resampled, y_train_resampled, X_test, y_test)
kn_accuracy, kn_error_rate, kn_ks_stat, kn_gini, kn_ypred = modeling("K-Nearest
Neighbors", kn_model, X_train_resampled, y_train_resampled, X_test, y_test)
lr_accuracy, lr_error_rate, lr_ks_stat, lr_gini, lr_ypred = modeling("Logistic
Regression", lr_model, X_train_resampled, y_train_resampled, X_test, y_test)
```

These configurations led to these results:

Summary of Model Classification Results							
Model	**Accuracy**	**Error Rate**	**Precision**	**Recall**	**F-1 score**	**K-S Statistic**	**Gini Coefficient**
Random Forest	82.09%	17.91%	0.85	0.80	0.82	1.00	1.00
Decision Tree	79.14%	20.86%	0.70	0.80	0.79	1.00	1.00
K-Nearest Neighbors	99.04%	0.96%	0.99	0.99	0.99	0.9981	0.9963
Logistic Regression	89.71%	10.29%	0.90	0.88	0.90	1.00	1.00

- The Random Forest model correctly predicted 82% of the test data, while roughly 18% of the predictions were incorrect. It achieved a perfect recall in classes 7 and 3 but was challenged with class 6 (0.26). All in all, it still has a high accuracy and is effective for classification tasks. This is suitable for tasks where moderate accuracy is acceptable since it can still capture complex patterns and handle non-linear relationships. but it will not always be reliable for highly imbalanced datasets.
- The Decision Tree model recorded a slightly lower accuracy (79%) than the Random Forest model. Some of the classes were classified perfectly, however, it struggled in others where it was unable to predict effectively. To conclude, the decision tree's performance may be skewed and may not be as reliable as other models. It is less reliable for heavy classification problems compared to other models. This is particularly useful for immediate, rule-based decision-making; however, it may not be the best independent model for large-scale predictions.
- The KNN model showed an exceptionally high accuracy of 99% and an extremely low error rate. All of its metrics are scored consistently high, implying that it has minimal classification and is considerably reliable. The

feasibility of this model in real-world applications relies on computation efficiency, wherein big datasets may slow inference, which makes it impractical for real-time application. It is more fitted for small to medium datasets.

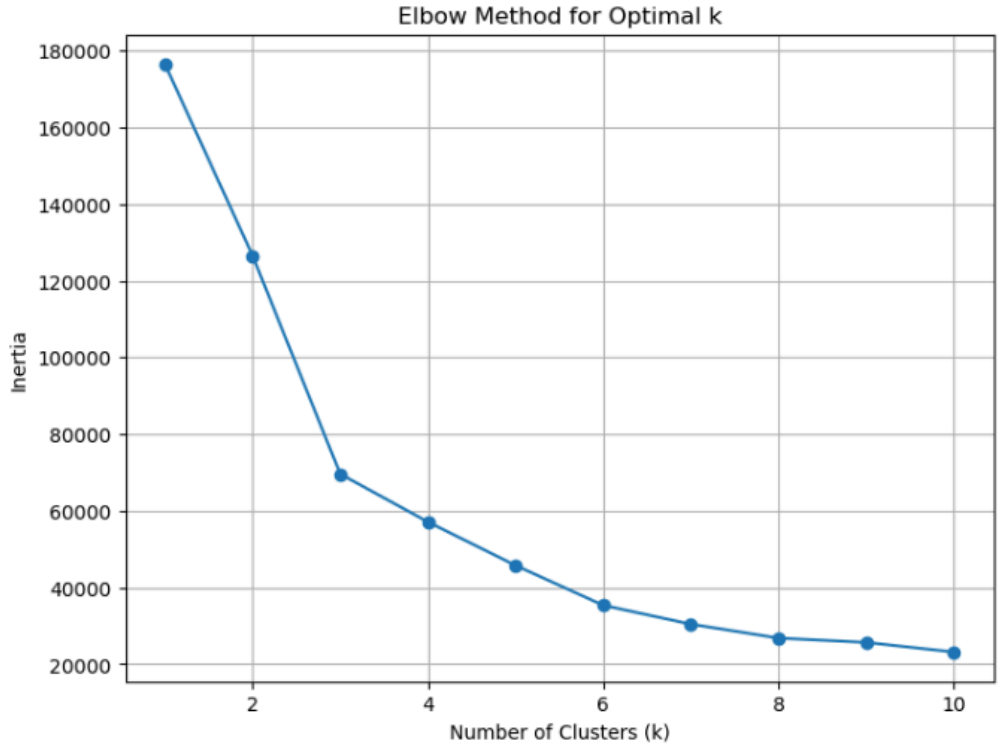
- The Logistic Regression model also achieved high accuracy (90%). It generally demonstrated robust performance due to its weighted average F-1 score, despite having low performance in certain classes. However, it may still struggle to accurately predict minority classes. This model performs well, but it may not be the best choice for very imbalanced data and complex patterns. This is particularly useful in aspects where explainability is essential, such as customer segmentation and fraud detection in business applications.

Model Evaluation with PCA and without PCA					
Model	**PCA**	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Random Forest	False	82.09%	0.85	0.80	0.79
	True	46.38%	0.47	0.45	0.43
Decision Tree	False	46.38%	0.70	0.80	0.73
	True	70.27%	0.73	0.70	0.69
K-Nearest Neighbors	False	99.04%	0.99	0.99	0.99
	True	99.56%	0.99	0.99	0.99
Logistic Regression	False	89.71%	0.90	0.88	0.89
	True	71.97%	0.70	0.65	0.64

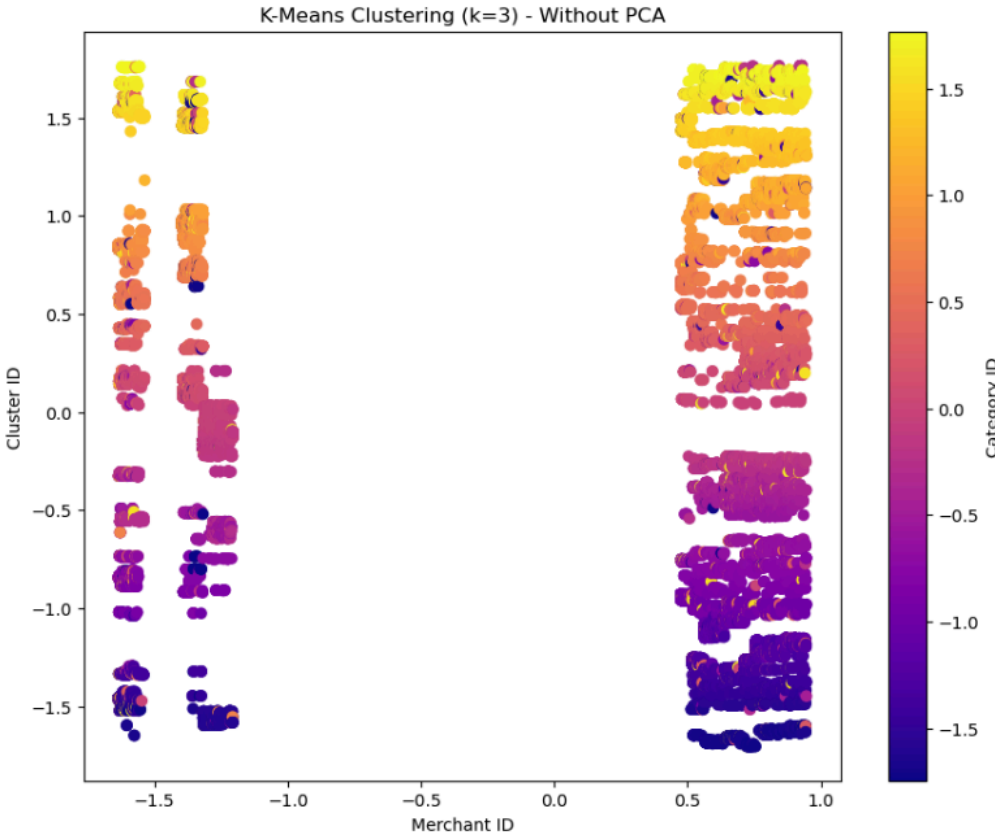
- It is evident that the KNN model performs the strongest in both with or without PCA, wherein both models are above 99%.
- Without PCA, Logistic Regression and Random Forest perform well with 89.71% and 82.09% accuracy scores, respectively. Meanwhile, the Decision Tree performs average without PCA too (>79%).
- It is observed that when PCA is applied to the model, the accuracy scores dropped significantly, except for the KNN model. This implies that PCA

might be discarding important features that are critical for classification in these models. PCA simplifies the data by converting it into fewer dimensions. In short, these models may depend heavily on feature interactions, impacting the accuracy of predictions.

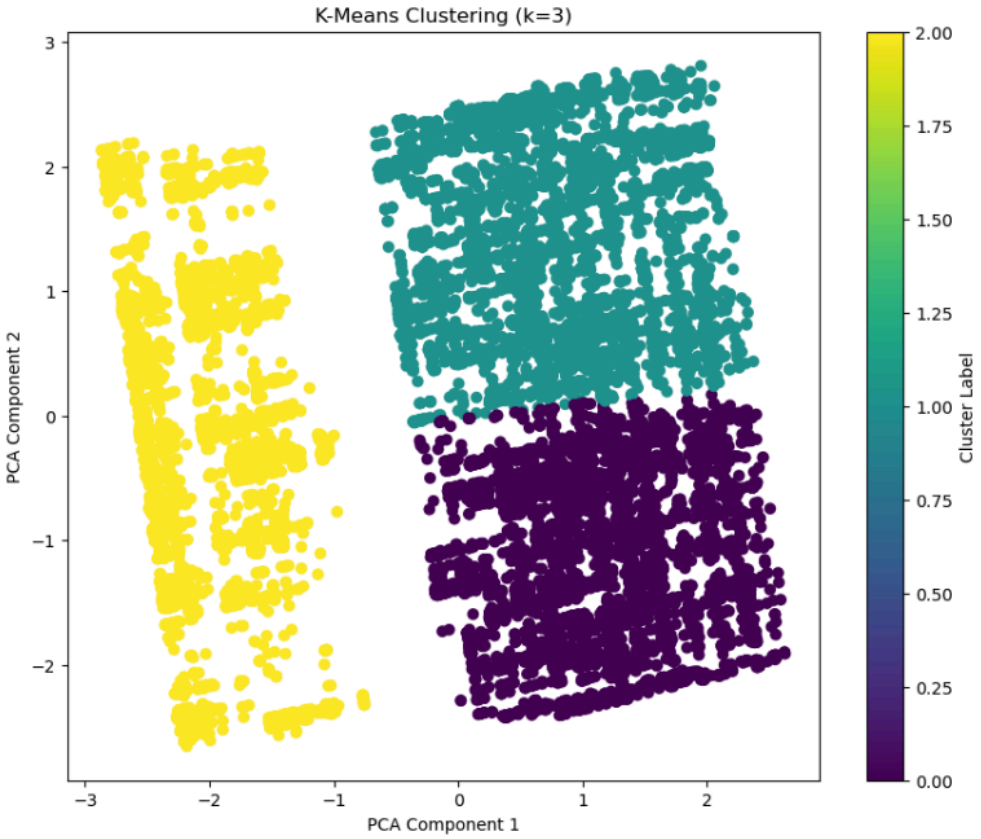
Proceeding with the clustering analysis, it is appropriate to test K-Means clustering through the Elbow method and Silhouette method to find an optimal number of clusters for the dataset (k).



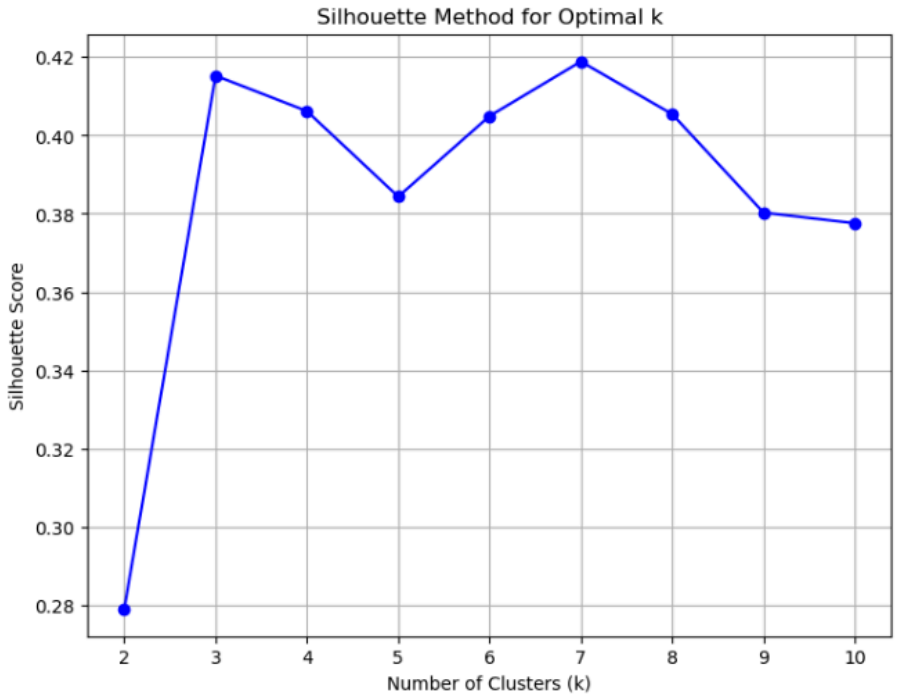
Starting off with the Elbow method, it illustrates that the optimal k is 3. It appears that the elbow point in the graph is 3 clusters or k=3 since it sharply declines at this point or the point where it starts to form an elbow. To visualize the clusters, both visualization with PCA and without PCA is done.



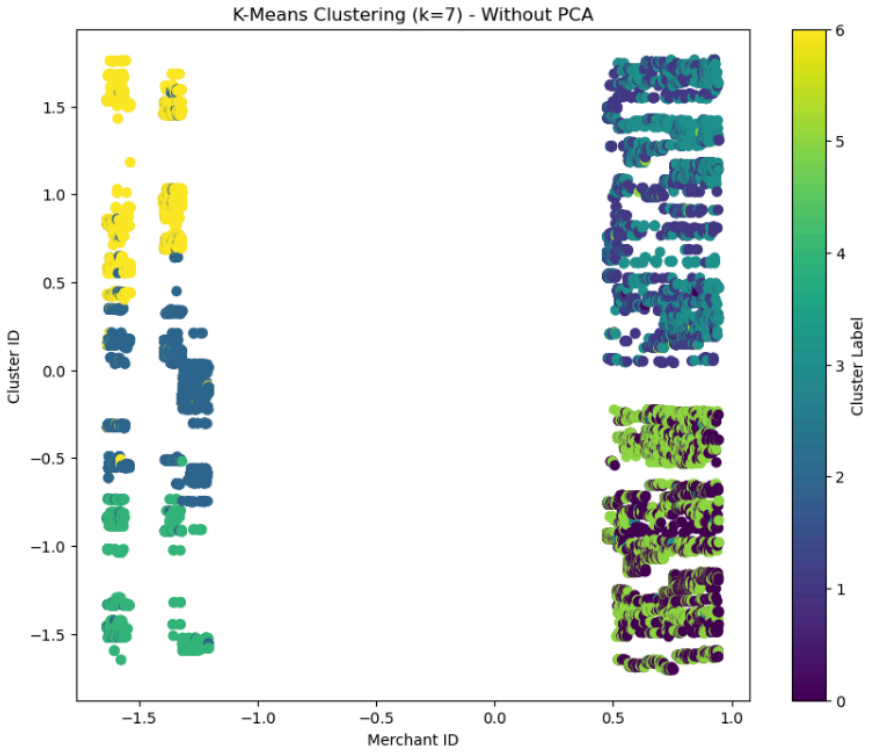
Without PCA, the features are directly plotted, which means the data is still in its original higher-dimensional space. It includes noise or irrelevant features that complicate the separation of clusters and are harder to visualize or analyze. Some features have more variance than others, but since there is no PCA applied, it is not prioritized. The complexity of the high-dimensional data is not reduced, which makes the clusters overall harder to distinguish from each other.

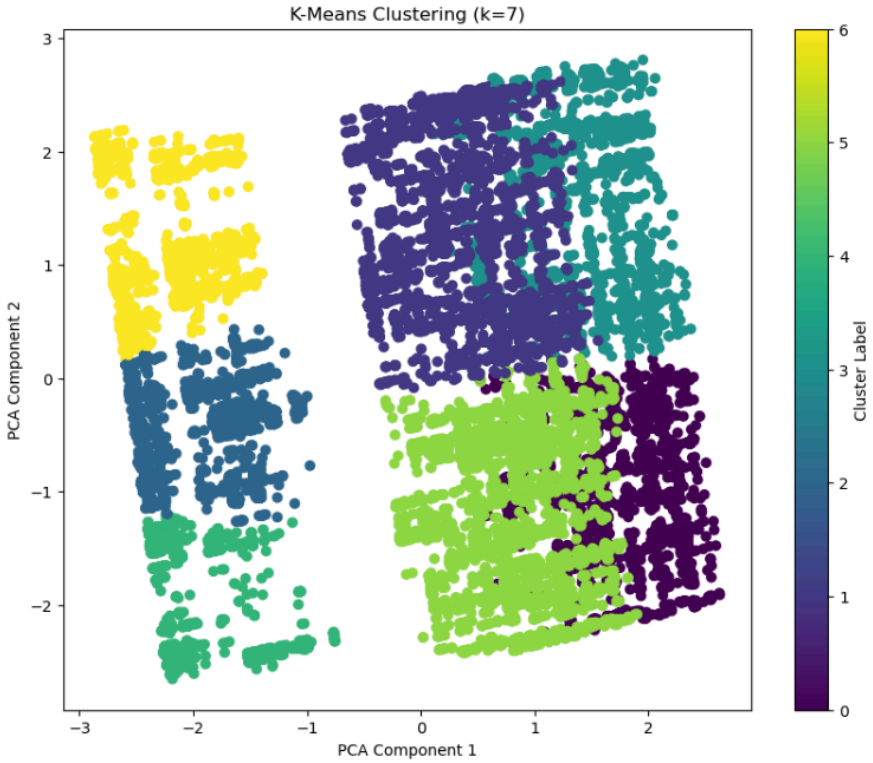


PCA captures the most variance in the data by identifying the principal components, which filters out the irrelevant features and noise and only highlights the key patterns. It is evident that it provides a clearer, organized separation between clusters, making it informative and visually appealing. In short, PCA presents a simplified version of the data by focusing on the important aspects and eliminating the irrelevant information.



On the other hand, the Silhouette method is executed, and the results produce a different optimal number of clusters. It is shown that the optimal k is 7. The highest Silhouette score or peak is at k=7, wherein it implies that 7 clusters is the best balance of cluster separation and cohesion.





Both graphs (Figure __ and __) show the same patterns similar to Elbow's method with and without PCA. It is observed that it generally overlaps or is harder to interpret when PCA is not applied to the clusters on both clustering analysis methods. Meanwhile, PCA is much easier to interpret and has cohesive clusters.

Program Design and Implementation

Software Tool Design

Supervised Classification Models

Models such as Random Forest, Logistic Regression, Decision Tree, and K-Nearest Neighbors are designed to predict the product categories based on features that are extracted. It follows a structure that includes data preprocessing, feature engineering, class imbalance handling, model training and evaluation, and performance enhancement methods.

1. **Data Preprocessing**

   Encoding categorical variables using LabelEncoder converts categorical data into numerical value, which makes it suitable for the models that require numerical input (GeeksforGeeks, 2018). Further, feature selection is executed to remove irrelevant features to reduce the feature space while improving the model's accuracy. Feature scaling using StandardScaler to standardize normalizes the numerical values, and it ensures that the performance of the models is better. To handle class imbalance, SMOTE is applied to the training set to mitigate class imbalances through the generation of synthetic samples to avoid biases toward majority classes.

2. **Model Training & Evaluation**

   The model trains several classification algorithms which allows for proper comparison and selection of the top-performing approach, such as Random Forest Classifier, Decision Tree Classifier, KNN Classifier, and Logistic Regression. The evaluation function, modeling(), evaluates the model performance through metrics beyond accuracy, which ensures an in-depth analysis. This includes accuracy, error rate, classification report, confusion matrices, Kolmogorov-Sminov (K-S) statistic, and Gini coefficient.

3. **Advanced Model Enhancements**

   The KNN model's robustness is assessed by adding controlled noise to training and testing sets. The program also applied Principal Component Analysis (PCA) to retain 95% variance while also reducing overfitting and improving

computational efficiency. The program tests each model's performance, with and without PCA, to identify its effect on classification accuracy.

This machine learning model can be described as a product classification system since it predicts the product's category based on the features. This model can be applied in business in many ways, including the automation of classifying new product listings to reduce manual work for managers. It will also assist businesses in tracking product categories and optimizing stock levels. Overall, it can improve customer satisfaction, operational efficiency, and decision-making of the administrative management.

Clustering Model

This model performs product categorization and segmentation with the help of PCA and K-means clusterings. The features are the following:

1. **Dimensionality Reduction & Cluster Segmentation**

   Principal Component Analysis (PCA) is used to reduce features while retaining important information, which allows better visualisation of product categories. The program also implements k-means clustering with the Elbow method and Silhouette Score to identify the optimal number of clusters.

2. **Cluster Analysis and Business Insights**

   The clusters are visualized through color-coded PCA plots for easier interpretation and analysis. This helps businesses recognize patterns, opportunities, and trends for better decision-making by identifying important features driving each cluster

Architecture

The model uses data science and machine learning libraries to support the whole pipeline, from data preprocessing to model training and evaluation.

1. **Data Handling and Processing**

   Pandas handles the manipulation of datasets, such as loading, cleaning, filtering, and transforming data, while Numpy is responsible for array manipulation and numerical operations.

2. **Feature Engineering & Preprocessing**

Sklearn.preprocessing, such as Label Encoder and Standard Scaler, scales numerical data and encodes categorical features to improve model performance. Additionally, sklearn.model_selection (train_test_split) splits the data into a particular ratio of training and testing sets. Imblearn.over_sampling corrects imbalanced datasets through SMOTE.

3. **Machine Learning Models**

   Libraries such as sklearn.ensemble, sklearn.tree, sklearn.neighbors, and sklearn.linear_model are responsible for implementing supervised machine learning models for classification tasks.

4. **Dimensionality Reduction & Clustering**

   Sklearn.decomposition and sklearn.cluster performs Principal Component Analysis (PCA) and K-Means Clustering, respectively.

5. **Model Evaluation and Statistical Analysis**

   Sklearn.metrics performs computation on the following evaluation metrics for machine learning models: accuracy, confusion matrices, classification reports, and silhouette scores. Meanwhile, scipy.stats (ks_2samp) conducts statistical tests like the K-S statistic.

6. **Data Visualization**

   Matplotlib.pyplot, and seaborn libraries are responsible for plotting graphs and clustering graphs to derive business insights.

Implementation

The model can be validated using the following metrics for classification and clustering, which serve as the test plan. Classification model validation metrics are for supervised learning models wherein the data are categorized based on their properties or traits into particular groupings. In order to check the accuracy of classifications, the model is measured with different metrics (GeeksforGeeks, 2023):

1. **Confusion Matrix** – It is a summary table of the classification algorithm's performance, wherein it consists of four metrics: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
    - True Positives (TP) are the instances where the model correctly predicted the positive class when it was truly present in the actual data.
    - True Negatives (TN) are the instances where the model correctly predicted the negative class when it was truly present in the actual data.
    - False Positives (FP) are the instances where the model incorrectly predicted the positive class when it was not really present in the actual data.
    - False Negatives (FN) are the instances where the model incorrectly predicted the negative class when it was actually a positive class.

Confusion Matrix Results

Random Forest		
True Positives (Diagonal Values)		
Category	**Class**	**Correctly Classified**
Highest Correct Predictions	4	955
	7	859
	3	443
Lowest Correct Predictions	6	121
	2	324
Misclassifications (Off-Diagonal Values)		
Misclassified Class	**Predicted As**	**Instances**

Class 0	8	83
Class 1	0	119
Class 2	1	243
	0	122
Class 6	1	263
	0	88
Class 9	2	133

Insights
It is observed that Class 6 instances (263) are frequently misclassified as Class 1 and 0. Similarly, Class 2 is highly confused with Class 1 and 0. Both Classes 2 and 6 have high misclassification rates. On the contrary, Class 3 has well-classified instances, Class 4 has high true positives with few errors, Class 5 has a small rate of misclassification, and lastly, Class 7 has strong separability and a lower confusion rate.

K-Nearest Neighbors		
True Positives (Diagonal Values)		
Category	**Class**	**Correctly Classified**
Highest Correct Predictions	4	1076
	7	859
	9	772
Lowest Correct Predictions	6	463
	3	436
Misclassifications (Off-Diagonal Values)		
Misclassified Class	**Predicted As**	**Instances**
Class 0	8	8

Class 1	2	2
	1	5
Class 2	0	1
	1	7
	0	5
Class 3	2	1
	4	2
	5	1
	9	4
Class 4	3	2
	5	1
	9	1
Class 5	4	9
Class 6	1	6
	2	1
	9	2
Class 9	2	4
	3	2
	4	1

Insights
It is evident that KNN performs great for Classes 4, 7, and 9 since it has exceptionally high correct classifications. Class 4 has the highest true positives, Class 7 has high accuracy, and Class 9 has a very strong classification. Meanwhile, Classes 6,3, and 2 have poor separability and show moderate confusion, particularly with neighboring classes. Overall, misclassifications are relatively low, which implies that KNN is still effective in distinguishing between classes and making accurate predictions.

Decision Tree

True Positives (Diagonal Values)		
Category	**Class**	**Correctly Classified**
Highest Correct Predictions	4	1081
	7	859
	9	729
Lowest Correct Predictions	6	472
	3	444
Misclassifications (Off-Diagonal Values)		
Misclassified Class	**Predicted As**	**Instances**
Class 5	4	693
Class 9	3	780
Insights Decision Tree performs great for most classes such as Class 0,1,2,6,7,8 and 4. These classes have very high accuracy with near-perfect classification. On the other hand, Class 5 is highly misclassified as Class 4 with 693 instances. Class 9 also suffers from strong misclassification into Class 3 with 780 instances. Despite the misclassification problems, the Decision Tree exhibits a strong overall performance in the dataset.		

Logistic Regression		
True Positives (Diagonal Values)		
Category	**Class**	**Correctly Classified**
Highest Correct Predictions	4	1069
	7	859
	8	729
Lowest Correct Predictions	6	345

	3	348
Misclassifications (Off-Diagonal Values)		
Misclassified Class	**Predicted As**	**Instances**
Class 2	6	124
Class 2	9	187
Class 3	9	88
Class 6	2	124
Class 9	2	206
Class 9	3	24
Insights Logistic Regression performs great on major classes wherein Classes 0,1,4,5,7,8, have very high accuracy, and Class 4 has the highest correctly classified with 1069 correct. However, the model struggles with differentiating between Classes 2,6, and 9, which leads to high rates of misclassification. Some classes were confused with other classes, such as Classes 6 and 9, which reduced their accuracy.		

2. **Accuracy** – It is a core metric that is used to assess the performance of classification models. It measures how many instances are correctly predicted in the dataset (both true positives and true negatives). It follows a formula $Accuracy = (TP + TN)/(TP + TN + FP + FN)$. . The metric, accuracy, is easy to interpret since it is expressed in percentages, which can be easily understood by both non-technical and technical stakeholders. It is known to be reliable as well in dealing with balanced datasets. However, it can be unreliable in imbalanced datasets and misleading in crucial applications since accuracy treats all types of errors equally.

   Accuracy Results

Model	Accuracy	Insights
Random Forest	82.09%	Random Forests have higher accuracy than Decision trees, which is expected because Random Forest unites multiple trees to improve generalization and

		reduce overfitting.
K-Nearest Neighbors	99.04%	KNN achieved the highest accuracy, which outperformed other models. This means that the dataset has patterns that are fitted for instance-based learning. It also suggests that predictions are more effective and accurate if the model is based on the closest training examples.
Decision Tree	79.14%	Decision Tree has the lowest accuracy score since it may not generalize as well as other models. However, it can still be considered moderate performance. This could be advantageous when it comes to interpretability, but the trade-off is lower accuracy compared to the rest of the models.
Logistic Regression	89.71%	Logistic Regression has exceptional performance, which implies that the dataset has a clear decision boundary that a linear model can manage effectively. The features of the dataset have strong separability between classes.

3. **Precision –** It is a core metric utilized to estimate the ratio of correctly identified positive instances to all the identified positive instances. This is crucial for minimizing false positives. In short, it provides insights into the model's ability to predict accurate positive instances. It follows a formula $Precision = TP/(TP + FP)$

Model	Precision	Observations

Random Forest	precision 0 0.63 1 0.45 2 0.71 3 1.00 4 1.00 5 0.85 6 1.00 7 1.00 8 0.89 9 1.00	It is observed that Classes 3,7, 6, and 9 have achieved a perfect precision (1.00), which means that the model has prevented misclassification. Classes 0, 2, and 5 can be considered moderate precision since it may have correctly determined positive cases, but it still makes errors. On the other hand, the model struggles greatly on Class 1 with 0.45 precision.
K-Nearest Neighbors	precision 0 0.99 1 0.99 2 0.99 3 0.99 4 0.99 5 1.00 6 0.97 7 1.00 8 1.00 9 0.98	Each of the classes has a very high precision score, nearly equal to 1.00. This means that almost all predictions made by the model are correct. It demonstrates exceptional accuracy and very few false positives. This very high precision may suggest that there is strong generalization in the dataset and that the model has learned the patterns of the data proficiently.
Decision Tree	precision 0 1.00 1 1.00 2 1.00 3 0.36 4 0.61 5 0.00 6 1.00 7 1.00 8 1.00 9 0.00	Classes 0, 1, 2, 6, 7, and 8 have a perfect precision of 1.00, which means it has well-defined patterns that can be easily classified. However, this may also show that there is a bias towards these classes while totally failing other classes such as 5 and 9. Classes 5 and 9 were never been correctly predicted by the model. This could be an indicator that there is an imbalance wherein Classes 5 and 9 are underrepresented. Adjusting class weights might help.

Logistic Regression	precision 0 1.00 1 1.00 2 0.56 3 0.94 4 1.00 5 1.00 6 0.82 7 1.00 8 1.00 9 0.66	It is evident that Classes 0, 1, 4, 5, 7, and 8 achieved a perfect precision (1.00), which means these classes are well-defined with clear feature distinctions. This makes them easy to classify correctly. On the other hand, Classes 2,6, and 9 have moderate precision wherein there can be room for improvement. Class 3 is high but not perfect, wherein there is still a few false positives. Since most classes have high precision, this indicates that the model is great at avoiding false positives.

4. **Recall (Sensitivity) –** It assesses the model's performance to correctly identify all positive cases. It follows a formula $Recall = TP/TP + FN$.

Model	Recall	Insights
Random Forest	recall 0 0.89 1 0.78 2 0.47 3 1.00 4 0.88 5 1.00 6 0.26 7 1.00 8 0.88 9 0.83	It is observed that Classes 3,5, and 7 have a perfect recall of 1.00, which means the model correctly identified all true instances of these classes. This means that the model consistently identifies these categories. Additionally, there is a high recall for Classes 0, 4, and 9, moderate recall for Classes 1 and 8, and very low recall for Class 6. Overall, the model is generally great at grasping correct instances but may struggle in specific classes.

K-Nearest Neighbors	recall 0 1.00 1 0.99 2 0.98 3 0.98 4 1.00 5 0.99 6 0.98 7 1.00 8 0.99 9 0.99	The recall values for KNN are consistently high, which suggests there are no or few false negatives for these classes. Overall, it has an exceptional recall performance, which is an excellent model choice for applications where recall is critical.
Decision Tree	recall 0 1.00 1 1.00 2 1.00 3 1.00 4 1.00 5 0.00 6 1.00 7 1.00 8 1.00 9 0.00	It is evident that it has perfect recall in most classes, however, it fails to correctly classify any instance from these categories (Classes 5 and 9). This indicates that there is highly inconsistent recall performance, which could be improved by adjusting hyperparameters or better tuning.
Logistic Regression	recall 0 1.00 1 1.00 2 0.62 3 0.78 4 0.99 5 1.00 6 0.73 7 1.00 8 1.00 9 0.71	A powerful performance is exhibited for most classes due to its perfect recall (0,1,5,7, and 8). However, it struggles with Class 3 due to potential misclassification issues. Classes 2, 6, and 9 can be tuned to increase moderate recall values.

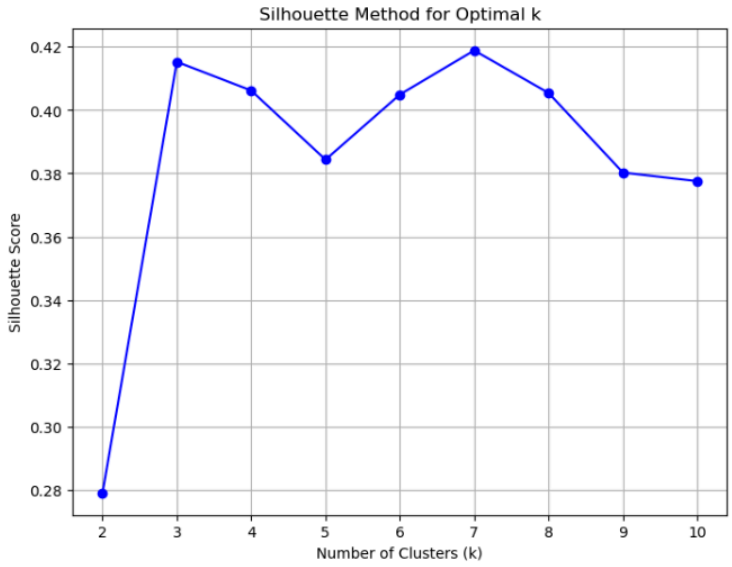
5. **F1-Score –** This metric combines both precision and recall into a single value, wherein it provides a balanced assessment of the overall model's performance. This is significant in handling class imbalance because this

metric considers both false positives and false negatives. It follows a formula $F1 - Score = 2x((Precision * Recall)/(Precision + Recall))$.

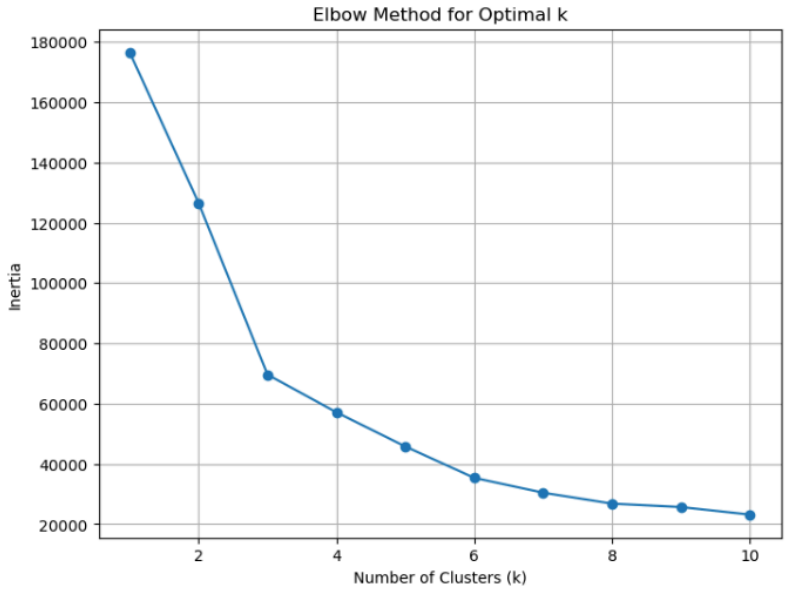
Model	Best F-1 Scores	Weakest F-1 Scores	Overall Performance
Random Forest	Classes 3 (1.00), 7 (1.00), and 9 (0.91)	Classes 1 (0.57), 2 (0.56), and 6 (0.41)	Random Forest is strong but is inconsistent in various classes (weighted F1 = 0.82). KNN has the best performance compared to the rest of the models, wherein the macro & weighted is F1=0.99. On the other hand, the Decision Tree may be unreliable due to its extreme variation wherein there is some perfect, and some failed; macro F1 = 0.7. Lastly, Logistic Regression is well-balanced but has weak spots in some categories; weighted F1 = 0.90.
K-Nearest Neighbors	Classes 0,1,2,6,7,8 (all 1.00)	Classes 5 (0.00), 9 (0.00), 3 (0.53), 4 (0.76)	
Decision Tree	Almost all classes (0.98 to 1.00)	None	
Logistic Regression	Classes 0,1,4,5,7,8 (all 1.00)	Clases 6 (0.77), 3 (0.85), 9 (0.68), 2 (0.59)	

In Clustering, validation techniques help evaluate the reliability and quality of clustering outcomes.

1. **Silhouette Score –** It is a validation technique that evaluates clusters without ground truth by measuring how similar a sample is to its own cluster in comparison to other clusters (silhouette_score, 2025). Some important parameters include metric, sample_size, and random state. The Silhouette Coefficient for a sample follows a formula $(b - a)/max(a, b)$.



2. **Elbow Method** – It is a validation technique that involves plotting the Within-Cluster Sum of Squares (WCSS) against the number of clusters (k) and determining the point where there is a formation of an “elbow” shape or the rate where it decreases sharply (GeeksforGeeks, 2019).



Evaluation and Ethical Considerations

Evaluation

To begin, the strengths of the program will be discussed. The program is composed of multiple techniques and analyses, including exploratory data analysis, clustering, and classification. These techniques are the foundation of machine learning and data insights. They allow analysts to extract meaningful insights, build models, and make informed decisions (Staff, 2025). Different classification and clustering models were compared, analyzed, and evaluated the outputs. The program also considered both technical and business perspectives since it covered merchant and product insights. Technical perspectives are crucial because they ensure the accuracy, reliability, and efficiency of the models. Meanwhile, business perspectives ensure that the analysis and models align with business objectives, such as improving product categorization, overall sales, and merchant performance. The project gave comprehensive business recommendations that can be understood by both technical and non-technical stakeholders. In the program, it was detected that there are overfitting issues wherein the initial accuracy is 99-100%. While it is an unfavorable result, recognizing it immediately enables timely adjustments, such as hyperparameter tuning. Hyperparameter tuning is essential for improving model's performance by selecting the optimal values of hyperparameters for a particular machine learning model (e.g., n_estimators, max_depth) (GeeksforGeeks, 2019). Along with that, the program implements corrective measures like increased regularization, limiting model complexity, and using SMOTE for mitigating class imbalances. Clustering techniques such as k-means clustering, the elbow method, and the silhouette method are critically evaluated. It was provided in the insights of specific applications or use cases for different models, which indicates which scenarios best fit each model. The business recommendations provided are actionable and straightforward.

Following the strengths, the weaknesses of the program is identified. First, the initial accuracy of 99-100% on machine learning models raises serious concerns. Overfitting can happen due to high variance and low bias, high model complexity, and size of the training data (GeeksforGeeks, 2017). Fortunately, the overfitting was resolved through the aforementioned adjustments. The program also lacks Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) scores conducted for model comparison (Agarwal, 2024). ROC plots the rate of true positives against false positives in order to measure the performance of a classification model. On the other hand, AUC is the area under the ROC curve, which is a measure to evaluate the binary classification model's performance. There are limitations in clustering analysis since there is a lack of mathematical justification or quantitative backing between the elbow method (k=3) and the silhouette method (k=7). Lastly, cross-validation isn't mentioned in

the analysis, except for the initial overfitting process. Overall, the program is stabilized and serves as a good foundation for data analysis and machine learning.

Some of the challenges encountered during development are data processing, clustering, and classification issues. Starting off with the data processing, the product title values in the dataset had inconsistencies when it came to misspellings, extra details, and word ordering. These inconsistencies made the results of frequency counts inaccurate. The initial method of counting exact matches led to an unrealistically low frequency value (90) instead of the real 2,540 occurrences found in Excel. The solution for this problem is to apply n-grams to capture multi-word patterns. N-grams are contiguous sequences of n words from a given sample of text (GeeksforGeeks, 2021). Identifying the optimal number of clusters was challenging as well since the silhouette method and elbow method yielded different optimal values (k=3 and k=7). Since the primary goal was to improve product categorization, overall sales, and merchant performance, k=3 was chosen since it gives well-organized and clear clusters. In regards to classification models, the models experienced difficulty in generalizing new data due to class imbalance and memorization of training data, which resulted in perfect accuracy across models. To avoid overfitting, many methods were utilized, such as increasing regularization, limiting model depth, applying SMOTH, and using PCA.

Ethical Implications

When machine learning models inherit patterns from data that indicate historical systemic prejudices, biases in Artificial Intelligence occur (mljourney, 2025). Any biases present in the dataset can lead to unfair outcomes since AI learns from pre-existing datasets. It may reinforce discrimination and misrepresent real-world situations. Biases can be addressed in many ways, including using representative and diverse datasets that include all demographic groups. In data analysis, these datasets will help avoid skewed insights. Another way is to implement fairness-aware machine learning models composed of bias detection and mitigation strategies. It is equally important to regularly audit AI systems and algorithms to make sure the models make unbiased decisions (mljourney, 2025). Therefore, it is important to have human oversight in AI-powered decision-making to correct for unintentional biases. This maintains a balance between human judgment and automation. It is crucial to consider transparency and explainability by developing models that give clear reasoning for their predictions without requiring specialized technical knowledge (Alagar, 2023). Best practices for transparency include documenting methodologies, assumptions, and data sources cohesively. It is also important to enable stakeholders to access information regarding the process of how data-driven conclusions are reached, which sheds light on the intricate factors that contribute to a prediction. Another ethical aspect is to ensure users are aware of how AI-driven decisions are made, which would foster trust and accountability.

Being accountable is an important quality for developers and organizations since it takes responsibility for the outcomes of their machine learning technologies. According to Mirror Review (2023), managing sensitive and personal data ethically requires strict adherence to privacy rules. In terms of privacy, ethical considerations include data collection for analysis purposes only, anonymization, acquisition of explicit informed consent, and compliance with GDPR to safeguard user information. To conclude, biases in machine learning and data analysis pose an ethical challenge since it amplify societal inequalities. This issue can be addressed through the curation of representative and diverse training datasets, regular assessment of models for biased outcomes, and enhancement in algorithmic decision-making. By cultivating accountability and fairness, inclusive and equitable outcomes of machine learning systems can be achieved instead of sustaining existing prejudices. Overall, mitigation of biases on machine learning and data analysis is not just a core technical necessity but a moral imperative to achieve ethical artificial intelligence.

Conclusion and Recommendations

Conclusion

In Exploratory Data Analysis, the dataset undergoes data cleaning and preprocessing through removing unnecessary space and checking for duplicated rows and missing values. A business question was formulated to extract in-depth analysis and insights from the dataset that would help in providing data-driven decisions to the company. Based on the results, 4-grams are useful for product titles because they effectively determine frequent combinations of terms, which highlight significant product attributes and categorization accurately. This method also improves upselling and product bundling by grouping relevant products together. Systematic product categorization is seen with a strong correlation between Category ID and Product ID. Merchants must focus on categories they highly concentrate on while also maintaining an optimal inventory for each item. It is important to build strong relationships with top-performing merchants and provide them with better promotional support. Along with that, the company can enhance product comparison by having detailed review ratings and side-by-side comparisons so customers make better shopping decisions. Clustering enhances the recommendation systems and optimizes the website interface for intuitive navigation.

At the initial evaluation of classification models, the results show that there is an exceptional performance on all models (~99%-100% accuracy). This investigation led to the conclusion that the models fail to generalize new data and must have memorized the training set due to the vast capacity to learn the data from it. Therefore, adjusting some parameters can help the model generalize better and lead to more realistic accuracy. It is found that increasing the strength of regularization and limiting the complexity of the model truly helps with overfitting. The Synthetic Minority Oversampling Technique (SMOTE) helps balance class distributions by creating synthetic examples of the minority class, which makes the model improve its performance and robustness. Lastly, KNN with PCA is the best model since it effectively captures local patterns in clustered data and reduces noise.

In clustering models, it is observed that the silhouette method has a different optimal k compared to the elbow method. The elbow method has an optimal k of 3, while the silhouette method is 7. Both depend on when to use them since the elbow method focuses on compactness and doesn't consider how well-seperated the clusters are. Meanwhile, the silhouette method evaluates both separation and cohesion. However, the elbow method seems more organized, while the silhouette method may have closer clusters. K=3 might have a good balance between simplicity and representativeness since it provides a well-defined and distinct cluster. This is ideal for broader product and merchant

categories. K=3 is better to use when developing more generalizable and manageable categories. K=7 reveals more detailed and refined insights that may allow for more precise segmentation, but it may seem to overlap. Since k=3 is more organized and the data focuses more on improving product categorization, sales strategy, and product performance, this may be preferable. On the other hand, k=7 can provide more detail and is willing to handle the complexity for deeper analysis.

Importance of ML and Data Analysis

Healthcare (Alam, 2024)

- Machine learning models can examine real-time and historical data to make predictions about potential disease outbreaks, including flu or malaria, which considers factors like humidity, temperature, and travel patterns. Predictive analytics helps improve patient survival rates and reduce the spread of diseases since it enables healthcare professionals to intervene as early as possible. In addition, machine learning assists in creating personalized medicine by analyzing medical history, lifestyle factors, and genetic information. It helps develop personalized treatments for patients to minimize negative reactions and to ensure better therapies. This makes the healthcare approach more precise when it comes to patient outcomes and reducing unnecessary treatments. Data analysis greatly supports Clinical Decision Support Systems (CDSS), which assist doctors in recommending treatments, diagnosing diseases, and avoiding medical errors to patients. These clinical systems utilize large datasets of patient records, medical literature, and medical case studies to provide data-driven and factual insights at the point of care.

Manufacturing (Shang and You, 2019)

- Data analysis and machine learning have improved its operations, cost-reduction, and data-driven decision-making. One of these applications is predictive maintenance, wherein it utilizes real-time and historical data from machinery to predict failures before they occur. Therefore, it extends the life span of equipment and reduces unplanned downtime and maintenance costs. Additionally, machine learning models are useful in examining sensor and visual data, which makes them very effective for quality control and defect detection. Through advanced image recognition algorithms, it has the ability to inspect products with a precision that is unable to be done by humans, which ensures that only high-quality products reach the market. Lastly, data analytics allows for better supply

chain management by optimizing inventory levels and predicting demand. Real-time insights help manufacturers prevent stockouts, reduce excess inventory, and improve supply chain efficiency.

Business (Mesropyan, 2023)

- According to Mesropyan (2023), machine learning improves business analytics through the automation of complex analytical processes. This is because traditional data analysis methods have difficulty keeping up with the growing complexity of business data. This is addressed by machine learning with the use of advanced algorithms to find patterns, correlations, and anomalies that might otherwise be neglected. Furthermore, it is also highlighted that machine learning has impacted customer engagement. Along with data analytics, businesses can personalize user experience through targeted advertising, recommendation systems, and predictive analytics, which boosts satisfaction and loyalty. Well-known companies such as Walmart, Tesla, and Uber have successfully implemented machine learning and data analytics to streamline operations, drive innovation, and improve efficiency.

Recommendations

It is recommended that Price Runner have a more effective systematic product categorization and recommendation systems, through clustering and classification, to enhance both the searchability of products and customer experience. It is important as well to encourage merchants to concentrate on their best categories and diversify their product offerings. It is recommended to create targeted marketing campaigns for top product categories like Fridge Freezers, Mobile Phones, and Washing Machines. The following classification models have their own distinct qualities, however, the KNN model is recommended due to its high precision. Other models can be used in certain situations:

- Random Forest for scenarios where high accuracy and balanced performance are required.
- Decision Trees where the model requires interpretability, like explaining varying decisions to stakeholders.
- Logistic Regression is used for simplicity and as a general-purpose model.

In clustering, use K=3 (or the elbow method) for broader categories like general inventory management, product categorization, and overall sales. On the other hand, K=7 (or the silhouette method) is used for in-depth analysis of product, merchant, and targeted marketing strategies. For future work, improvements can be done such as fine-tuning K-means hyperparameters, using advanced clustering algorithms (e.g., DBSCAN, hierarchical clustering), experimenting with gradient boosting, using stacked models, and hyperparameter optimization (e.g., Grid Search, Bayesian Optimization). Aside from improving machine learning models, implementing cross-validation, developing real-time insights, and testing more optimal K values. These recommendations will enhance the accuracy and interpretability of the analyses and software tool.

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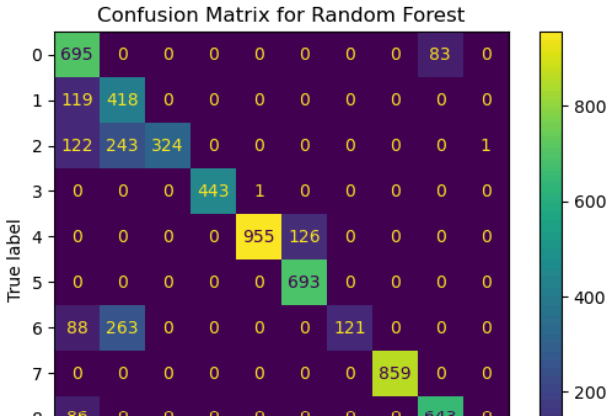
Appendix

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Evaluating Random Forest Model
Accuracy of Random Forest: 82.09%
Error Rate of Random Forest: 17.91%
Classification Report for Random Forest:
              precision    recall  f1-score   support

           0       0.63      0.89      0.74       778
           1       0.45      0.78      0.57       537
           2       0.71      0.47      0.56       690
           3       1.00      1.00      1.00       444
           4       1.00      0.88      0.94      1081
           5       0.85      1.00      0.92       693
           6       1.00      0.26      0.41       472
           7       1.00      1.00      1.00       859
           8       0.89      0.88      0.88       729
           9       1.00      0.83      0.91       780

    accuracy                           0.82      7063
   macro avg       0.85      0.80      0.79      7063
weighted avg       0.86      0.82      0.82      7063

Confusion Matrix for Random Forest:
[[695   0   0   0   0   0   0   0  83   0]
 [119 418   0   0   0   0   0   0   0   0]
 [122 243 324   0   0   0   0   0   0   1]
 [  0   0   0 443   1   0   0   0   0   0]
 [  0   0   0   0 955 126   0   0   0   0]
 [  0   0   0   0   0 693   0   0   0   0]
 [ 88 263   0   0   0   0 121   0   0   0]
 [  0   0   0   0   0   0   0 859   0   0]
 [ 86   0   0   0   0   0   0   0 643   0]
 [  0   0 133   0   0   0   0   0   0 647]]
```



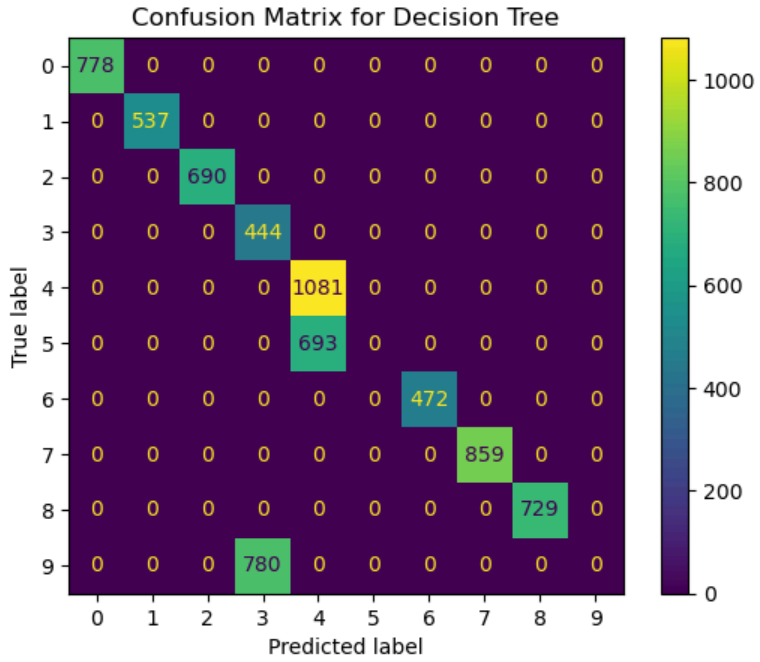
```
K-S Statistic for Random Forest: 1.0000, p-value: 0.0000
Gini Coefficient for Random Forest: 1.0000

Evaluating Decision Tree Model
Accuracy of Decision Tree: 79.14%
Error Rate of Decision Tree: 20.86%
Classification Report for Decision Tree:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       778
           1       1.00      1.00      1.00       537
           2       1.00      1.00      1.00       690
           3       0.36      1.00      0.53       444
           4       0.61      1.00      0.76      1081
           5       0.00      0.00      0.00       693
           6       1.00      1.00      1.00       472
           7       1.00      1.00      1.00       859
           8       1.00      1.00      1.00       729
           9       0.00      0.00      0.00       780

    accuracy                           0.79      7063
   macro avg       0.70      0.80      0.73      7063
weighted avg       0.69      0.79      0.72      7063

Confusion Matrix for Decision Tree:
[[ 778    0    0    0    0    0    0    0    0    0]
 [   0  537    0    0    0    0    0    0    0    0]
 [   0    0  690    0    0    0    0    0    0    0]
 [   0    0    0  444    0    0    0    0    0    0]
 [   0    0    0    0 1081    0    0    0    0    0]
 [   0    0    0    0  693    0    0    0    0    0]
 [   0    0    0    0    0    0  472    0    0    0]
 [   0    0    0    0    0    0    0  859    0    0]
 [   0    0    0    0    0    0    0    0  729    0]
 [   0    0    0  780    0    0    0    0    0    0]]
```

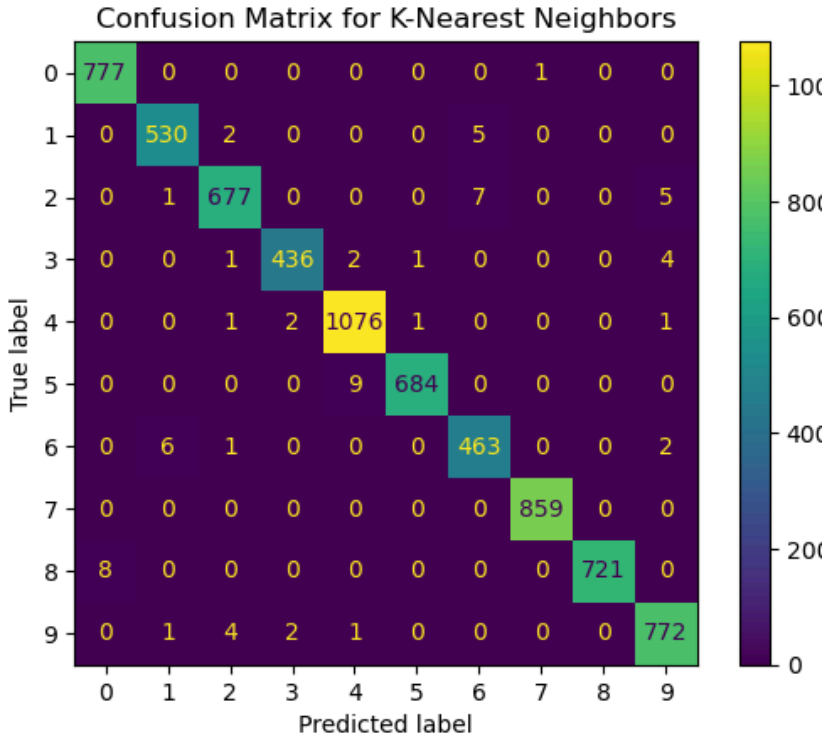


K-S Statistic for Decision Tree: 1.0000, p-value: 0.0000
Gini Coefficient for Decision Tree: 1.0000

Evaluating K-Nearest Neighbors Model
Accuracy of K-Nearest Neighbors: 99.04%
Error Rate of K-Nearest Neighbors: 0.96%
Classification Report for K-Nearest Neighbors:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	778
1	0.99	0.99	0.99	537
2	0.99	0.98	0.98	690
3	0.99	0.98	0.99	444
4	0.99	1.00	0.99	1081
5	1.00	0.99	0.99	693
6	0.97	0.98	0.98	472
7	1.00	1.00	1.00	859
8	1.00	0.99	0.99	729
9	0.98	0.99	0.99	780
accuracy			0.99	7063
macro avg	0.99	0.99	0.99	7063
weighted avg	0.99	0.99	0.99	7063

```
Confusion Matrix for K-Nearest Neighbors:
[[ 777    0    0    0    0    0    0    1    0    0]
 [   0  530    2    0    0    0    5    0    0    0]
 [   0    1  677    0    0    0    7    0    0    5]
 [   0    0    1  436    2    1    0    0    0    4]
 [   0    0    1    2 1076    1    0    0    0    1]
 [   0    0    0    0    9  684    0    0    0    0]
 [   0    6    1    0    0    0  463    0    0    2]
 [   0    0    0    0    0    0    0  859    0    0]
 [   8    0    0    0    0    0    0    0  721    0]
 [   0    1    4    2    1    0    0    0    0  772]]
```



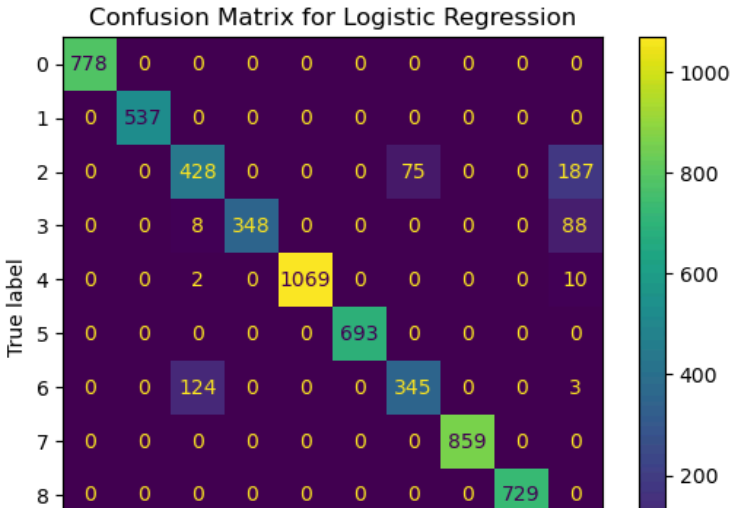

```
K-S Statistic for K-Nearest Neighbors: 0.9981, p-value: 0.0000
Gini Coefficient for K-Nearest Neighbors: 0.9963
```

```
Evaluating Logistic Regression Model
Accuracy of Logistic Regression: 89.71%
Error Rate of Logistic Regression: 10.29%
Classification Report for Logistic Regression:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       778
           1       1.00      1.00      1.00       537
           2       0.56      0.62      0.59       690
           3       0.94      0.78      0.85       444
           4       1.00      0.99      0.99      1081
           5       1.00      1.00      1.00       693
           6       0.82      0.73      0.77       472
           7       1.00      1.00      1.00       859
           8       1.00      1.00      1.00       729
           9       0.66      0.71      0.68       780

    accuracy                           0.90      7063
   macro avg       0.90      0.88      0.89      7063
weighted avg       0.90      0.90      0.90      7063

Confusion Matrix for Logistic Regression:
[[ 778    0    0    0    0    0    0    0    0    0]
 [   0  537    0    0    0    0    0    0    0    0]
 [   0    0  428    0    0    0   75    0    0  187]
 [   0    0    8  348    0    0    0    0    0   88]
 [   0    0    2    0 1069    0    0    0    0   10]
 [   0    0    0    0    0  693    0    0    0    0]
 [   0    0  124    0    0    0  345    0    0    3]
 [   0    0    0    0    0    0    0  859    0    0]
 [   0    0    0    0    0    0    0    0  729    0]
 [   0    0  206   24    0    0    0    0    0  550]]
```




```
K-S Statistic for Logistic Regression: 1.0000, p-value: 0.0000
Gini Coefficient for Logistic Regression: 1.0000
```